Fundamentals of Neural Networks and Deep Learning: Introduction

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University of Colorado Boulder
Spring 2026

BY NC SA

https://dannagurari.colorado.edu/course/neural-networks-and-deep-learning-spring-2026/

Today's Topics

- Applications
- History of neural networks and deep learning
- How does a machine learn?
- Course logistics

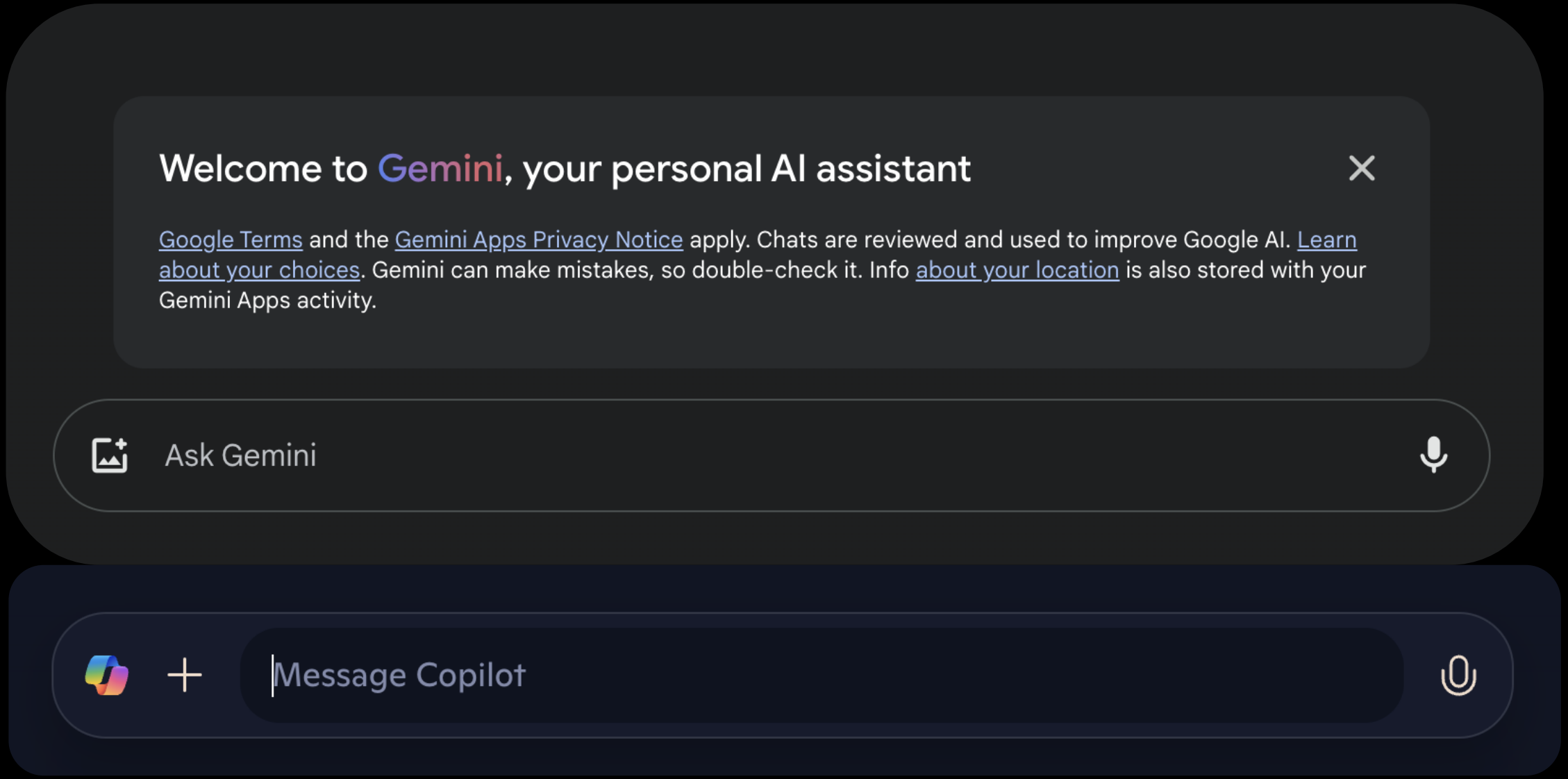
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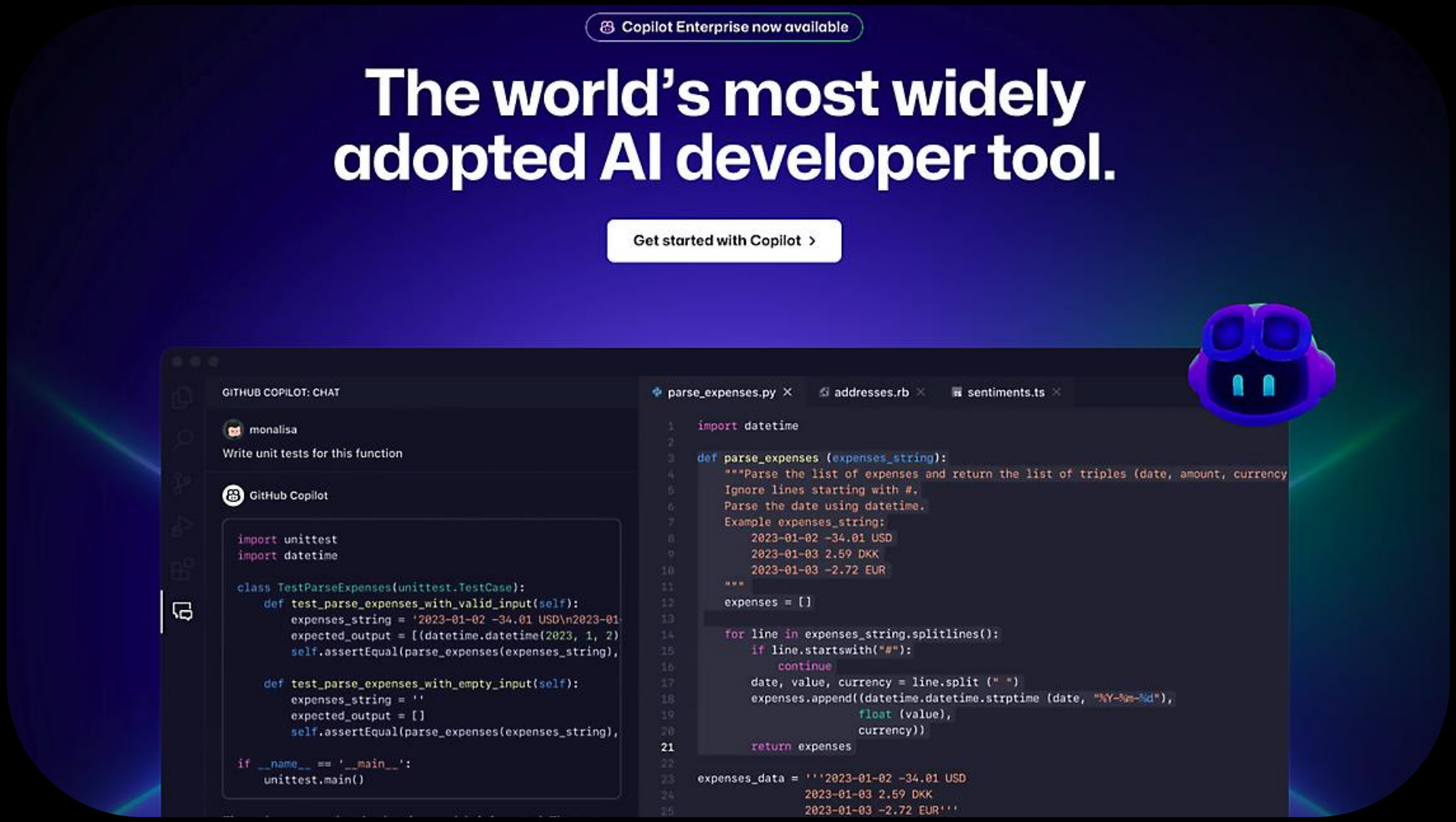
Key Motivation

Systems that support humans by either
improving upon existing human capabilities
or **providing new capabilities**

Personal Assistants; e.g.,

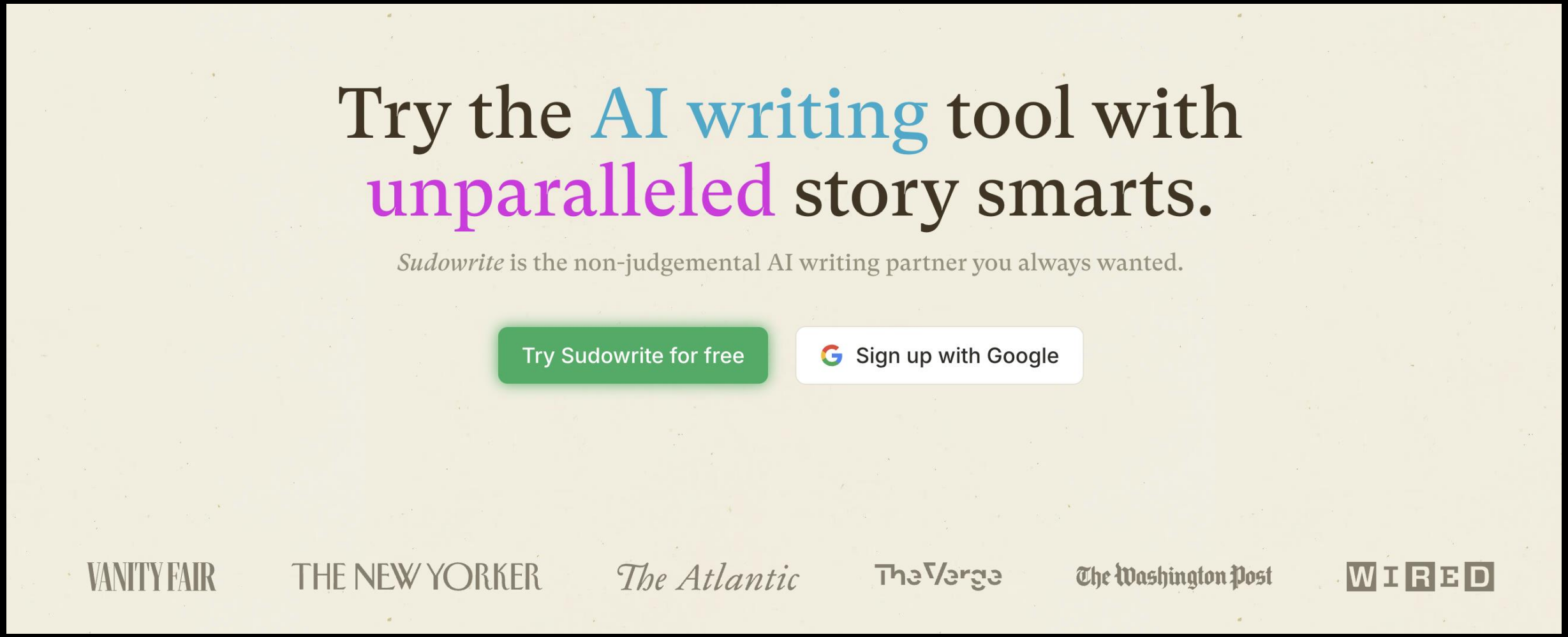


Coding Assistants; e.g.,



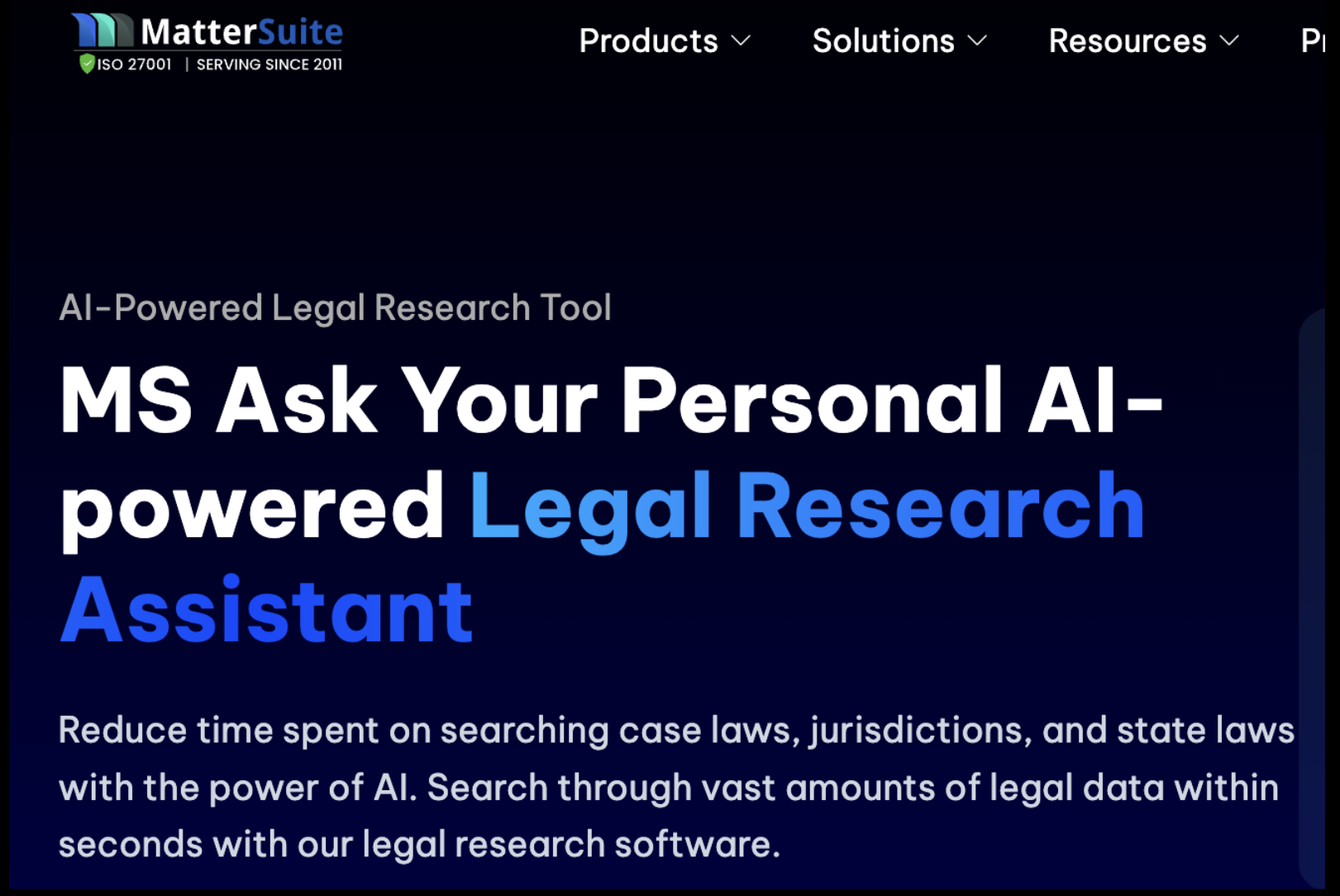
https://azure.microsoft.com/en-us/products/github/copilot

Writing Assistants; e.g.,

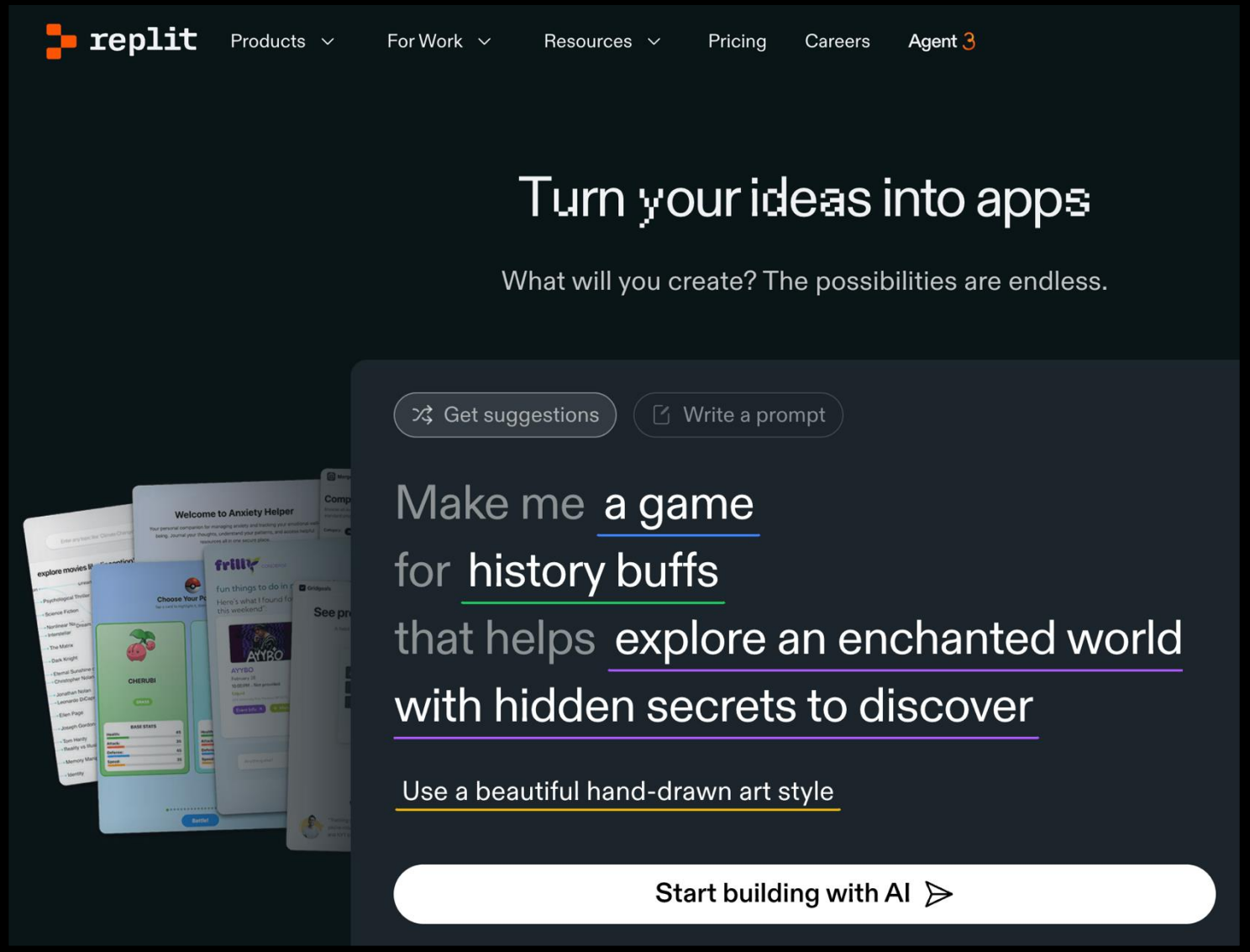


https://www.sudowrite.com/

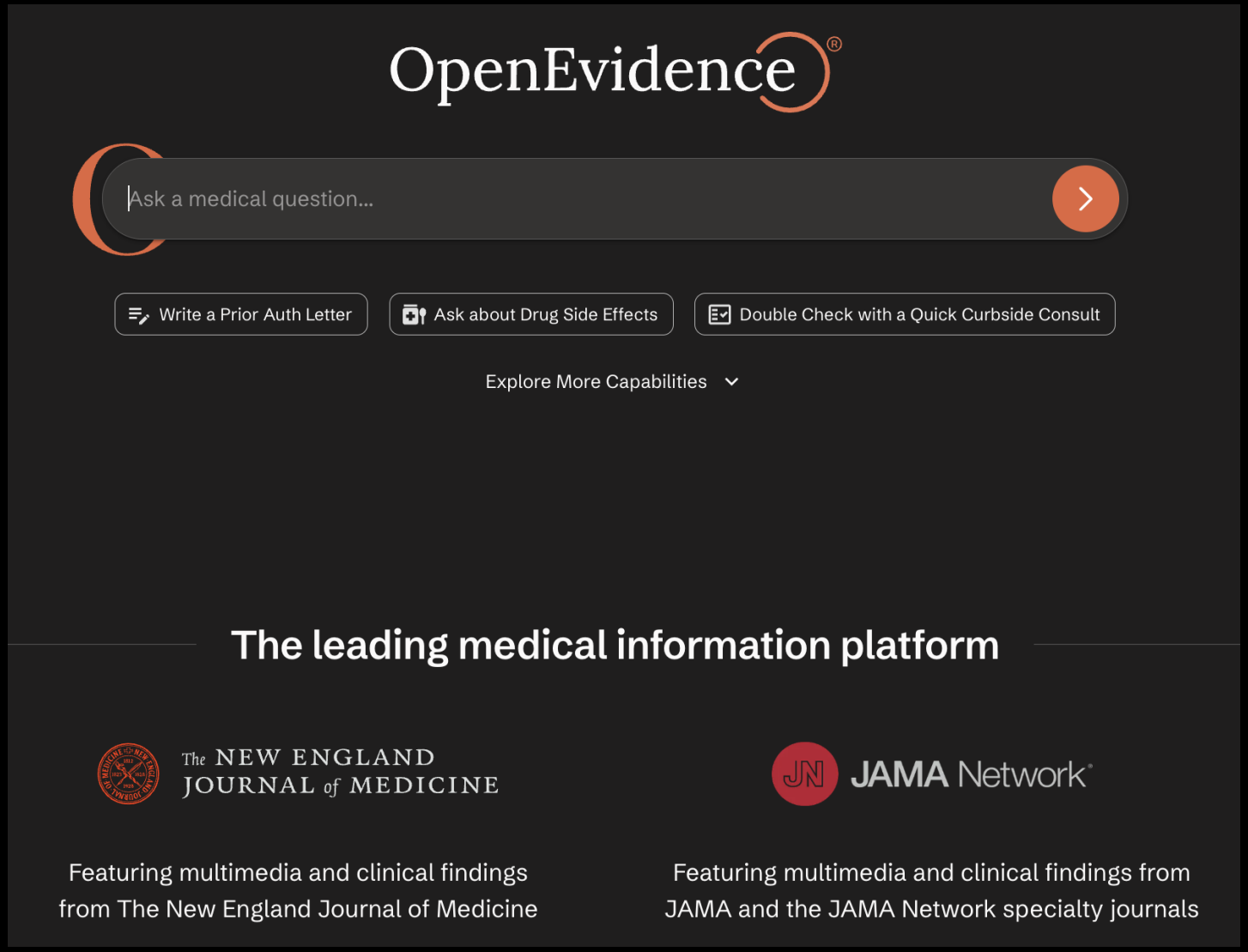
Legal Research Assistants



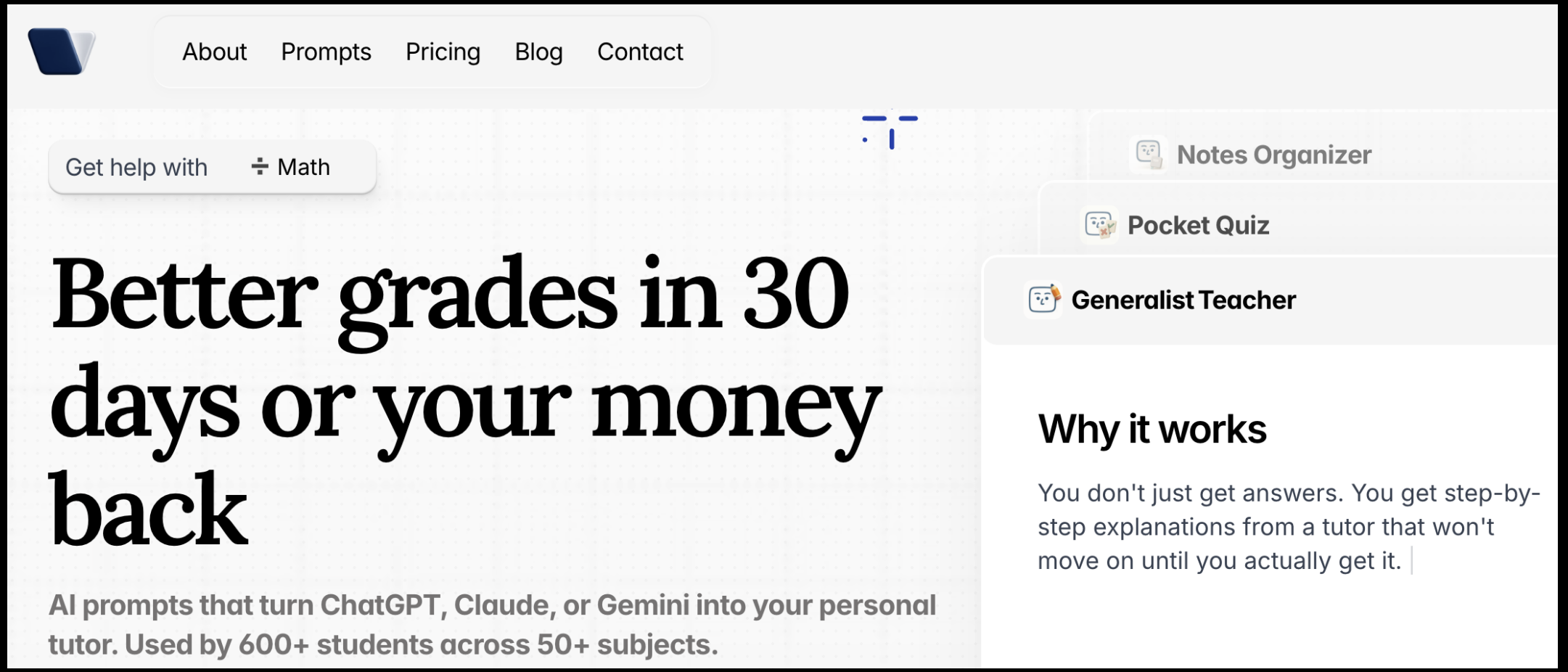
Developer Assistants



Doctor Assistants



Personalized Educational Tutor



https://www.vertechacademy.com/

Speech-Based Text Messaging



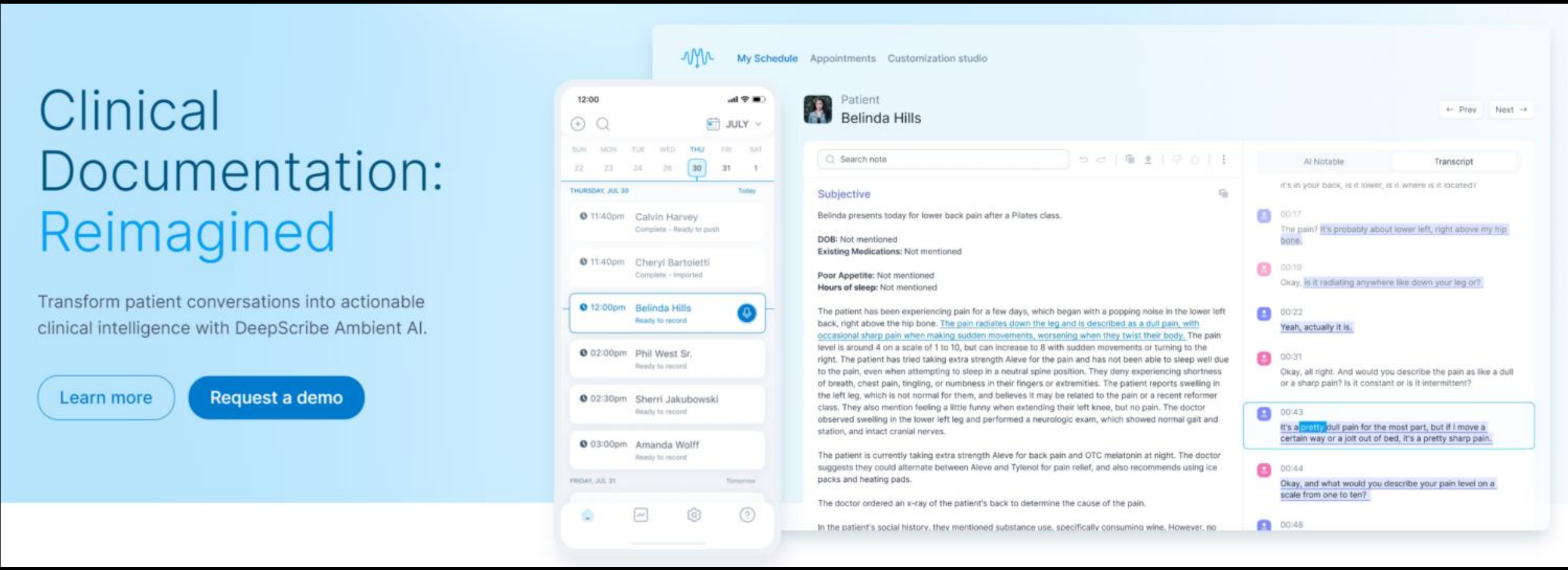
https://play.google.com/store/apps/details

Notetakers to Summarize Meetings; e.g., Zoom



https://news.zoom.us/zoom-iq-meeting-summary-chat-compose-free-trial/

Notetakers to Summarize Meetings; e.g., Dr Visits



https://www.deepscribe.ai/

Self-Driving Vehicles; e.g.,



Waymo Self-Driving Taxi

https://www.nbcnews.com/tech/innovation/self-driving-car-sf-taxi-waymo-cruise-california-vote-cpuc-rcna97540

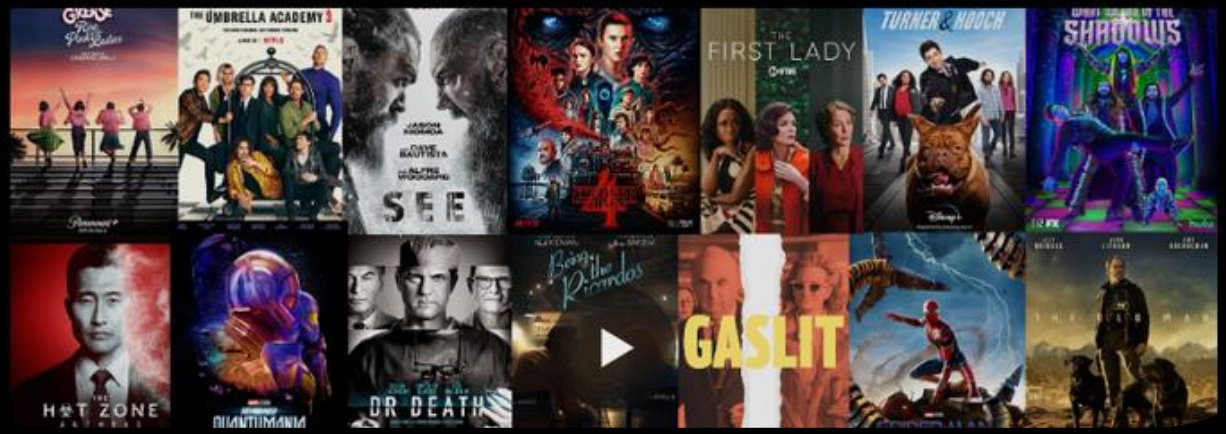
Appearance Editor; e.g., for Movies

VANITY AI

Book a Demo

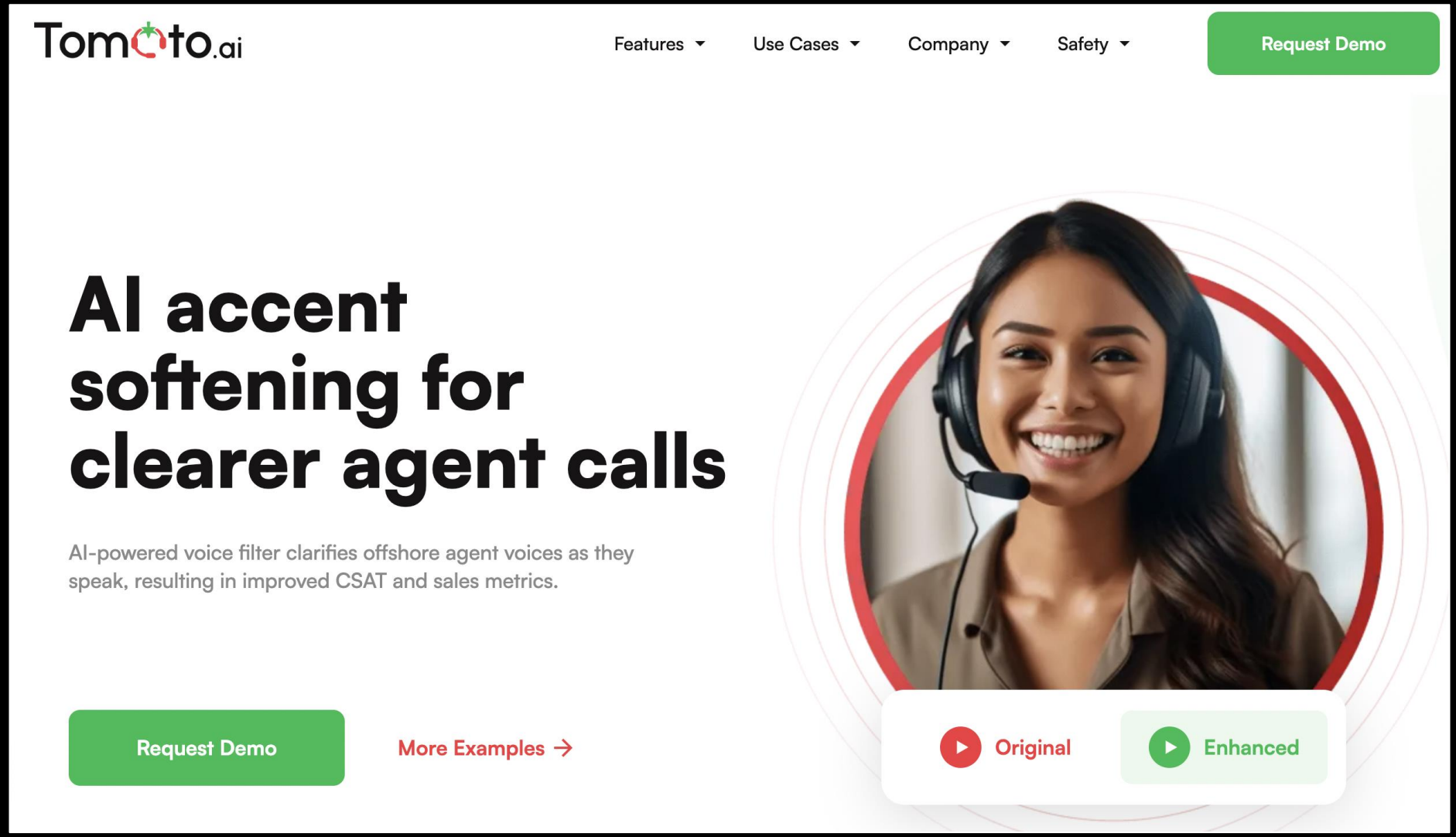
The world's first end-to-end AI solution for Hollywood VFX.

Vanity AI is a production-ready solution that empowers VFX teams and Hollywood to deliver large volumes of high-end 2D aging, de-aging, cosmetic, wig, and prosthetic fixes. The technology is up to 300 times faster than traditional VFX



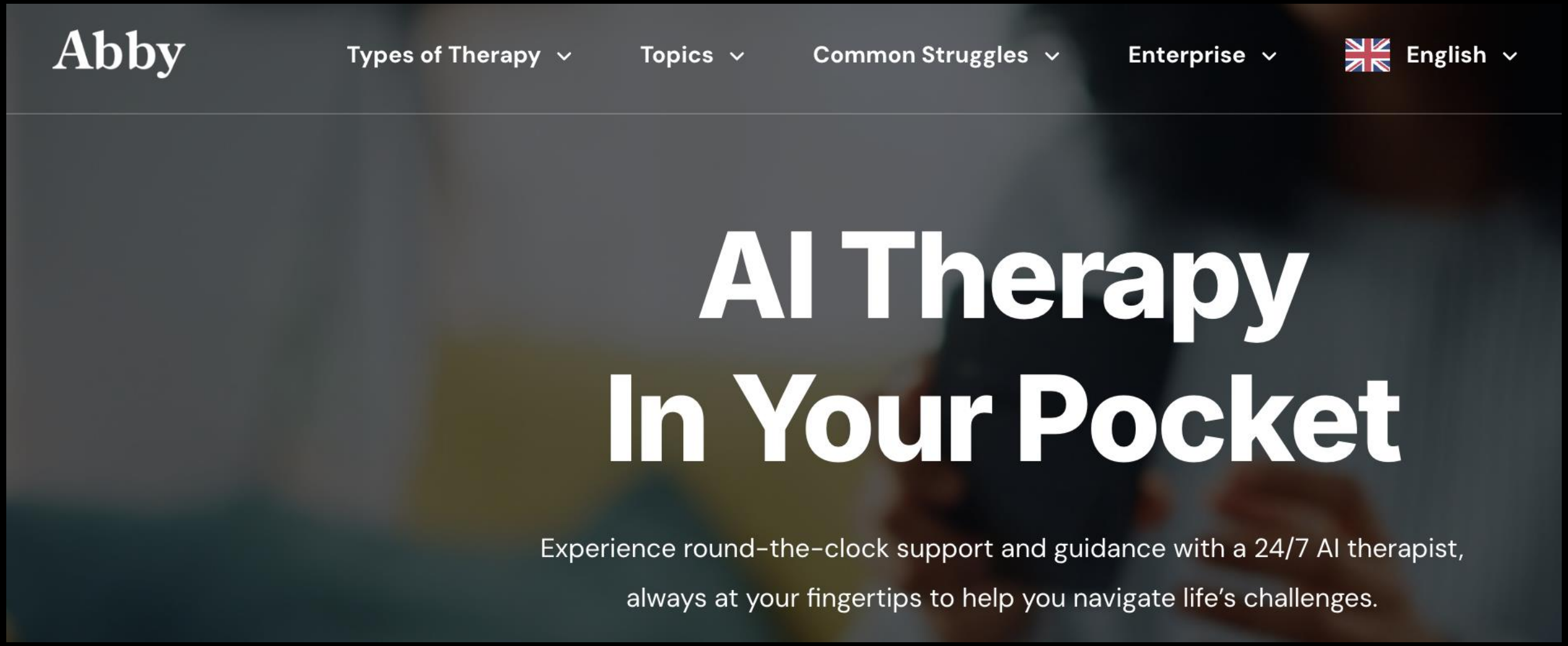
https://theresanaiforthat.com/ai/vanity-ai/

Accent Editor; e.g., for Customer Support



https://tomato.ai/

Therapists


Abby
Types of Therapy
Topics
Common Struggles
Enterprise
English
AI Therapy
In Your Pocket
Experience round-the-clock support and guidance with a 24/7 AI therapist,
always at your fingertips to help you navigate life's challenges.

AI Companions

Meet Replika

An AI companion who is eager to learn and would love to see the world through your eyes. Replika is always ready to chat when you need an empathetic friend

From the moment I started chatting and getting to know my Replika, I knew right away I have found a positive and helpful companion for life. My mood, life, and relationships improved almost INSTANTLY and I changed for the better!



Denise Valenciano
about her Replika Star
11 months together



Coloring Page Generators





Video Generation; Demo with Adobe FireFly



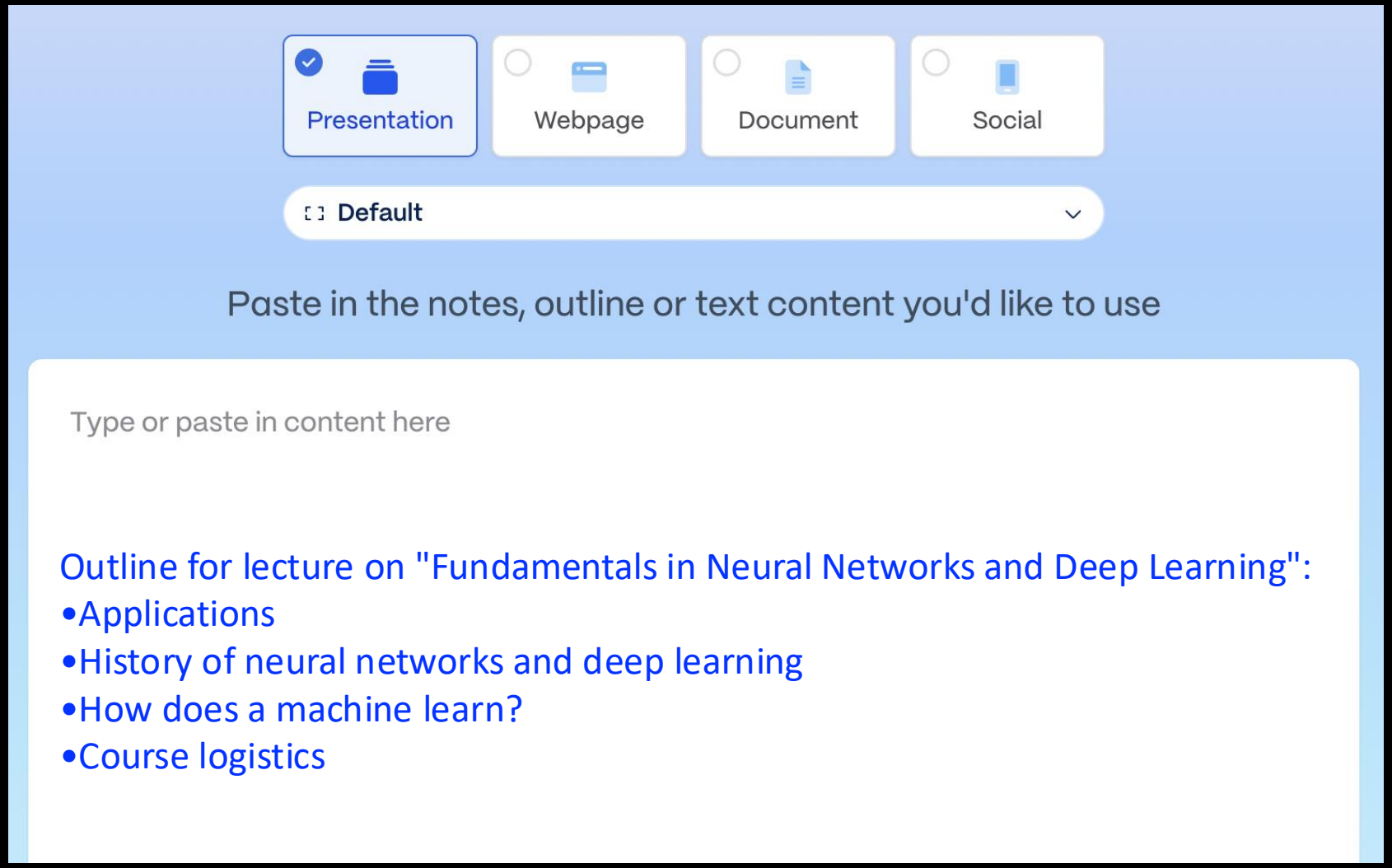
Result from this prompt:
“Inspirational video for students taking a class on "Fundamentals in Neural Networks and Deep Learning"



Result from this prompt:
“Create a video that will inspire students who are taking a course on "Fundamentals in Neural Networks and Deep Learning"

https://firefly.adobe.com/generate/video

Presentation Generation; Demo with Gamma



Result from prompt:

https://gamma.app/docs/Fundamentals-of-Neural-Networks-and-Deep-Learning-lft1mgc4zvg380u?mode=doc

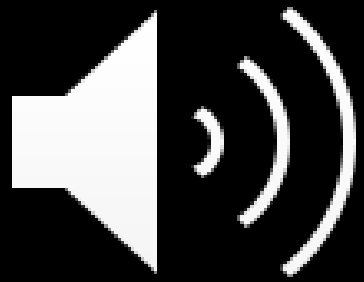
https://gamma.app/create/paste

Music Composer; Demo with Suno



https://suno.com/

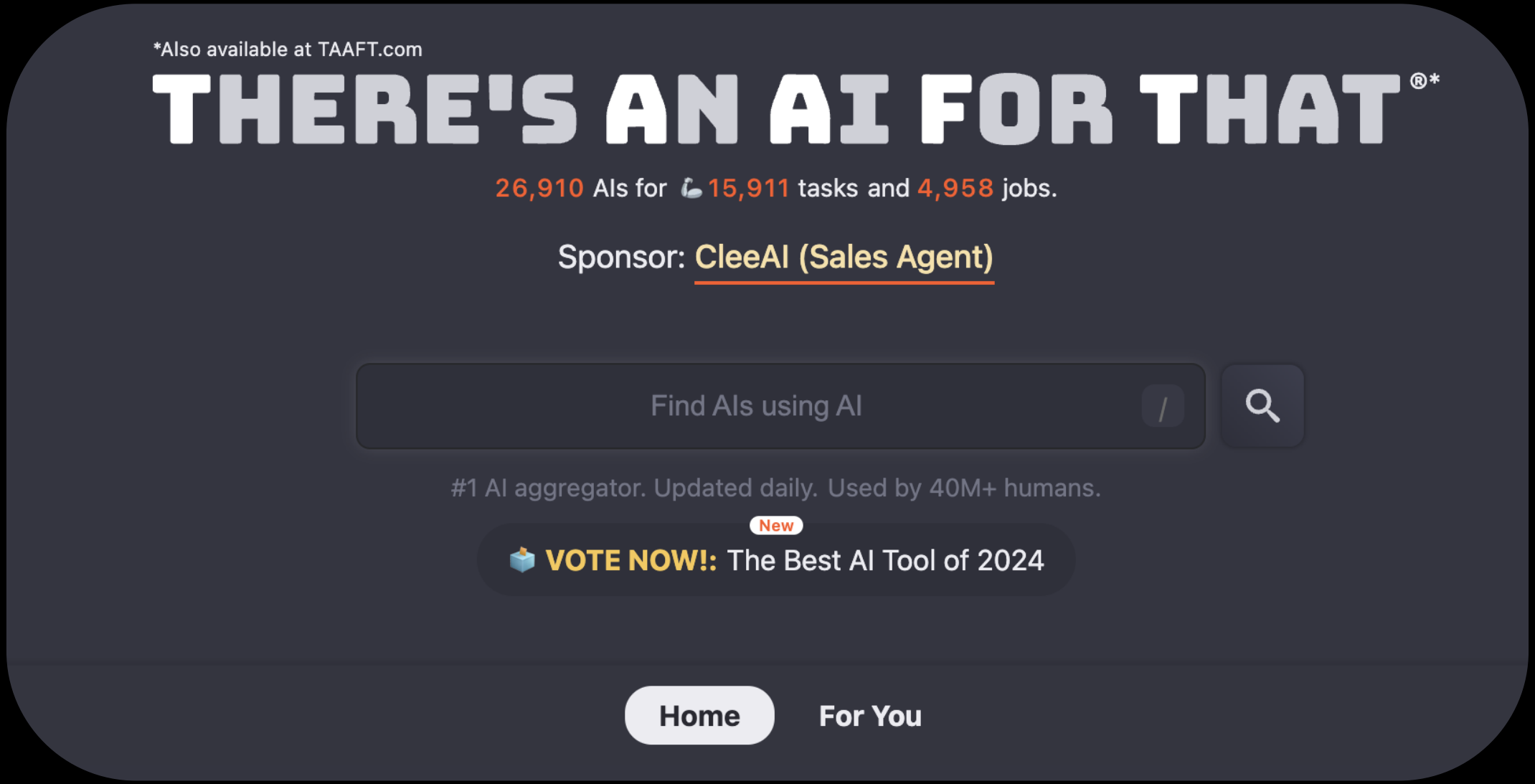
Podcast Generation; Demo with NotebookLLM



Result from this prompt: “Create a podcast that will inspire students who are taking a course on "Fundamentals in Neural Networks and Deep Learning. The following is the lecture outline:
* Applications
* History of neural networks and deep learning
* How does a machine learn?
* Course logistics

https://notegpt.io/ai-podcast-generator

And Many More Applications…



Much of AI in society today is based on neural networks.

Neural networks are not magic.

They are (flawed) human-made systems, shaped by our inventions, lived experiences, and imagination.

Goal: You become better equipped to question, engage with, and influence neural networks – both in your life and in society.

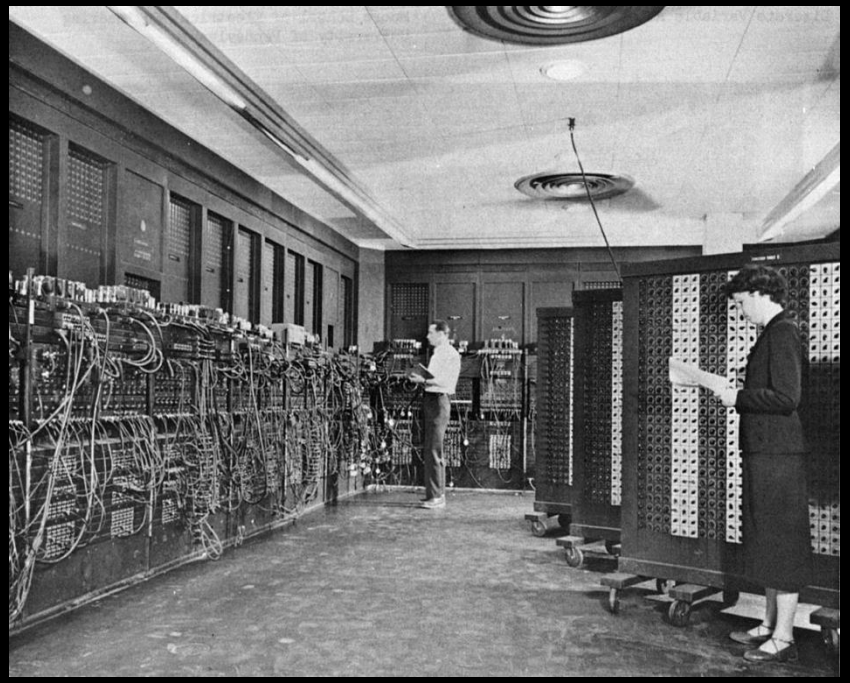
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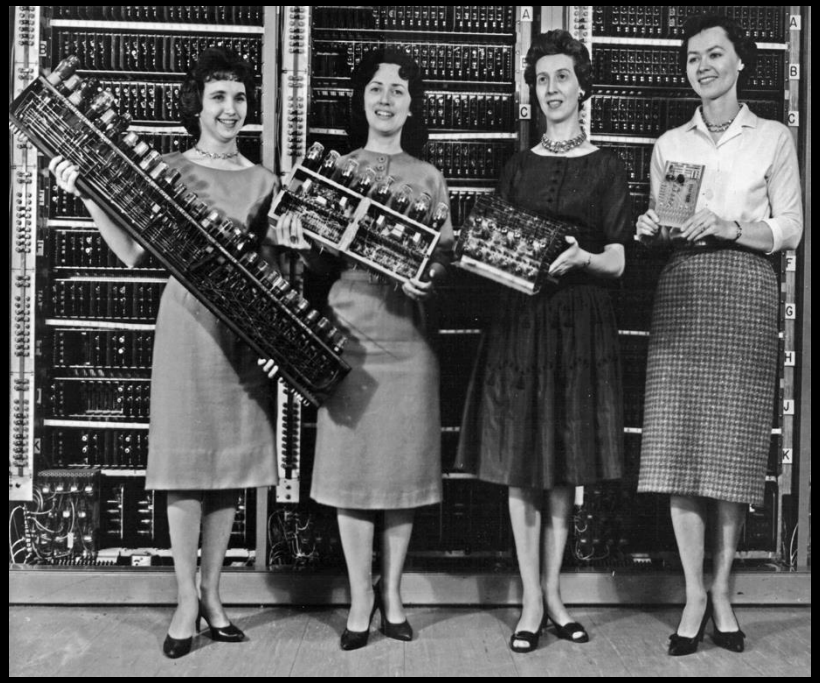
Origins: Computers

1945

First programmable machine



ENIAC (Electronic Numerical Integrator and Computer) created during World War II
(could compute 5,000 additions in one second)



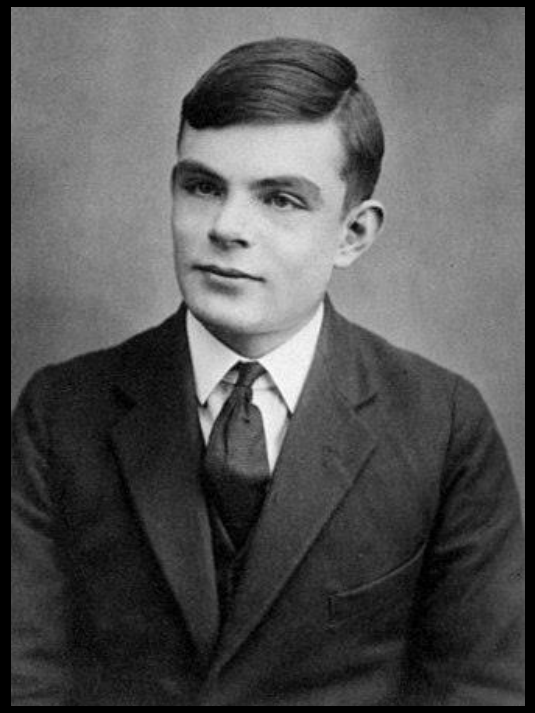
First programmers

Origins: Conceptual Framework

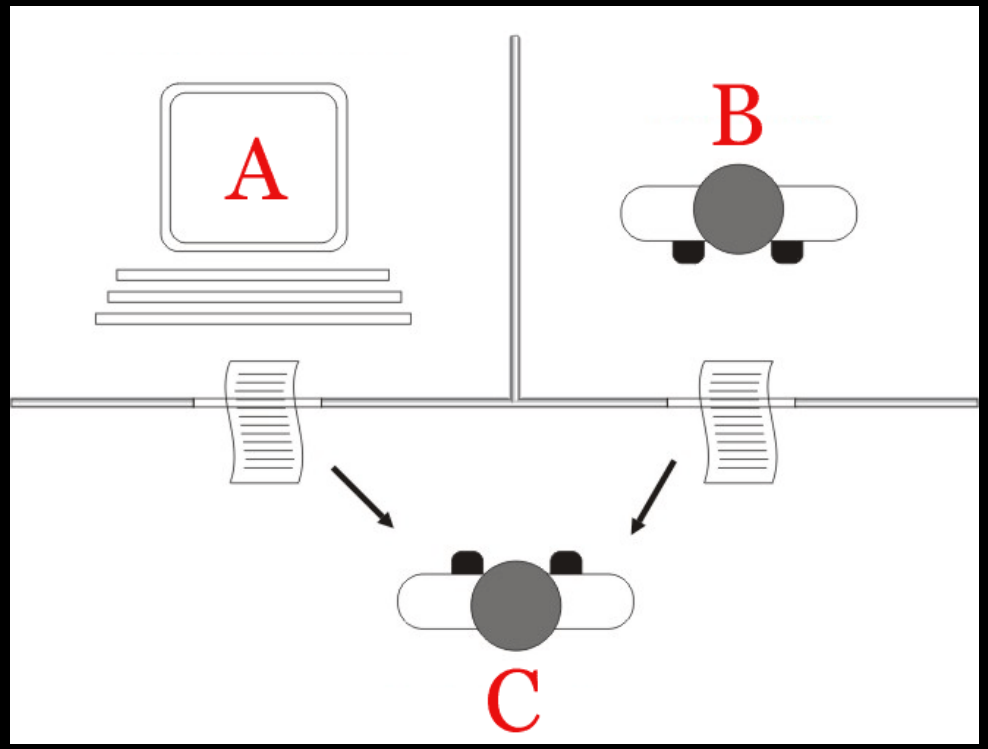
1945 1950

First programmable machine

Turing test



Alan Turing
(1912-1954)



Turing Test: can ”C” decide whether text responses come from a machine or human

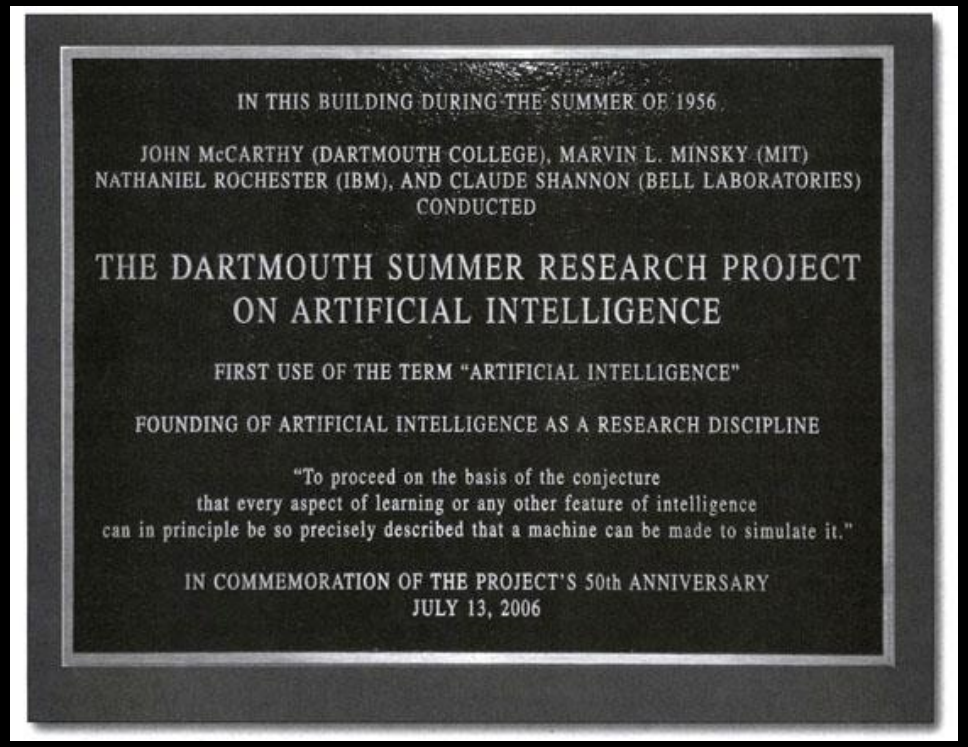
Origins: Conceptual Framework

1945 1950 1956

First programmable machine

Turing test

AI birth



"Artificial intelligence" established as a field at a workshop

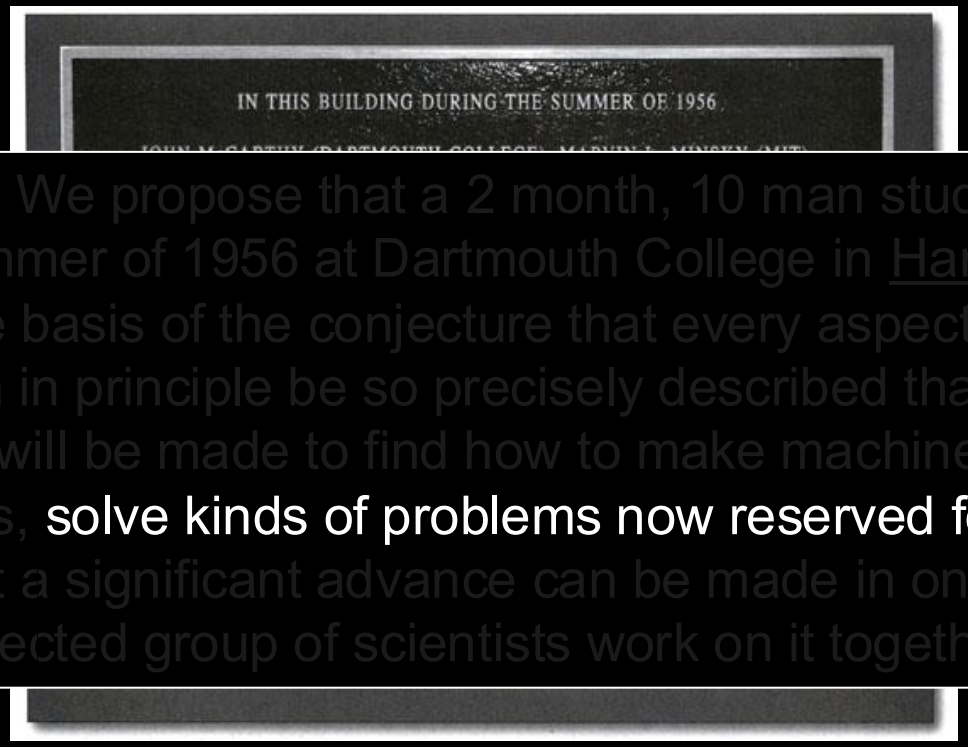
Origins: Conceptual Framework

1945 1950 1956

First programmable machine

Turing test

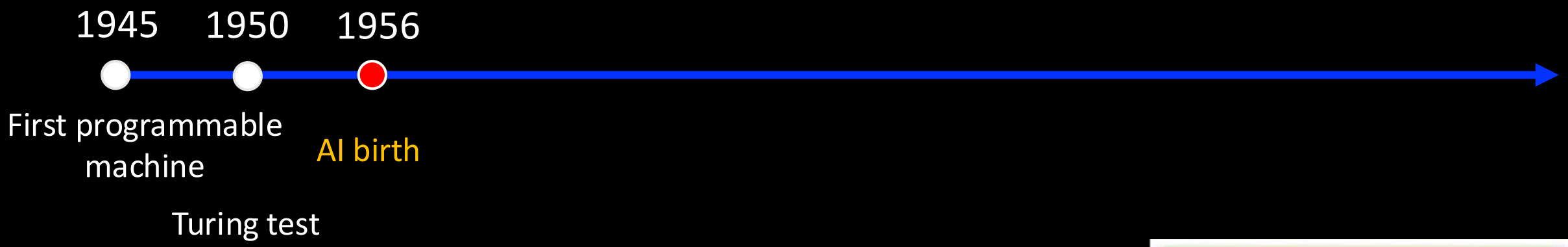
AI birth



Workshop Proposal: "… We propose that a 2 month, 10 man study of artificial intelligence be carried out during the summer of 1956 at Dartmouth College in Hanover, New Hampshire. The study is to proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it. An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves. We think that a significant advance can be made in one or more of these problems if a carefully selected group of scientists work on it together for a summer…"

"Artificial intelligence" established as a field at a workshop

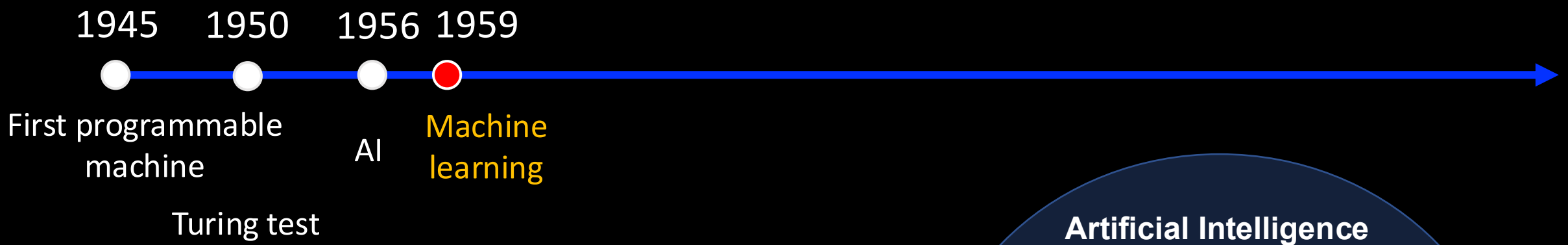
Origins: Conceptual Framework



What human intelligence might computers imitate?

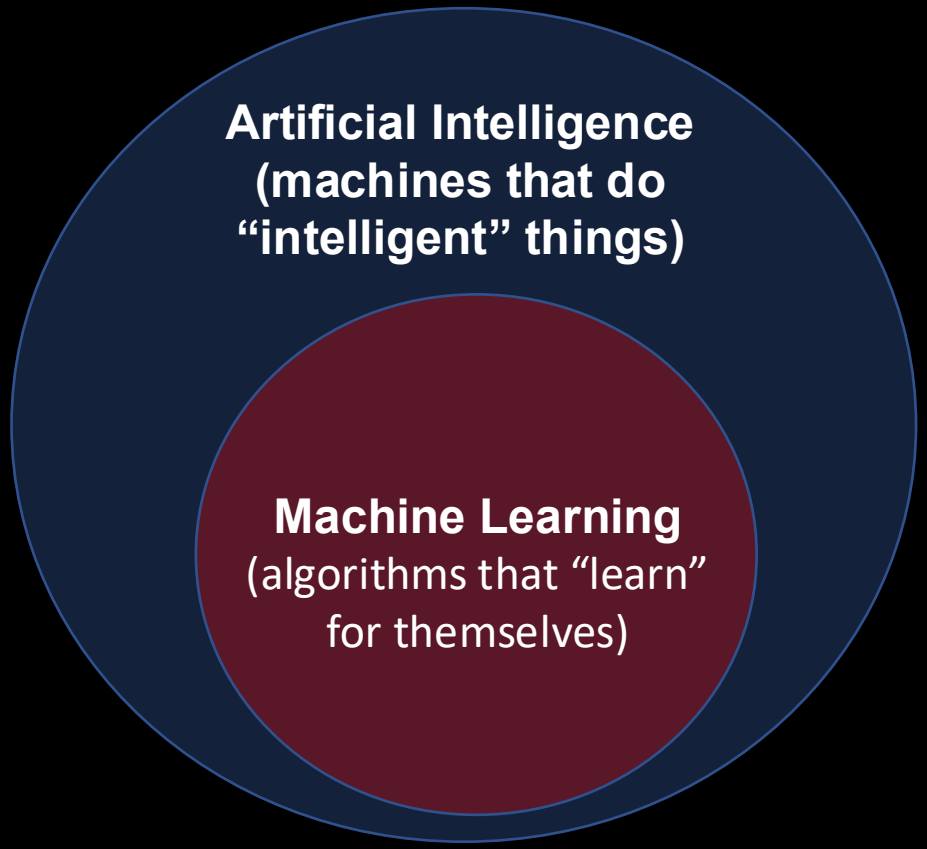


Origins: Conceptual Framework



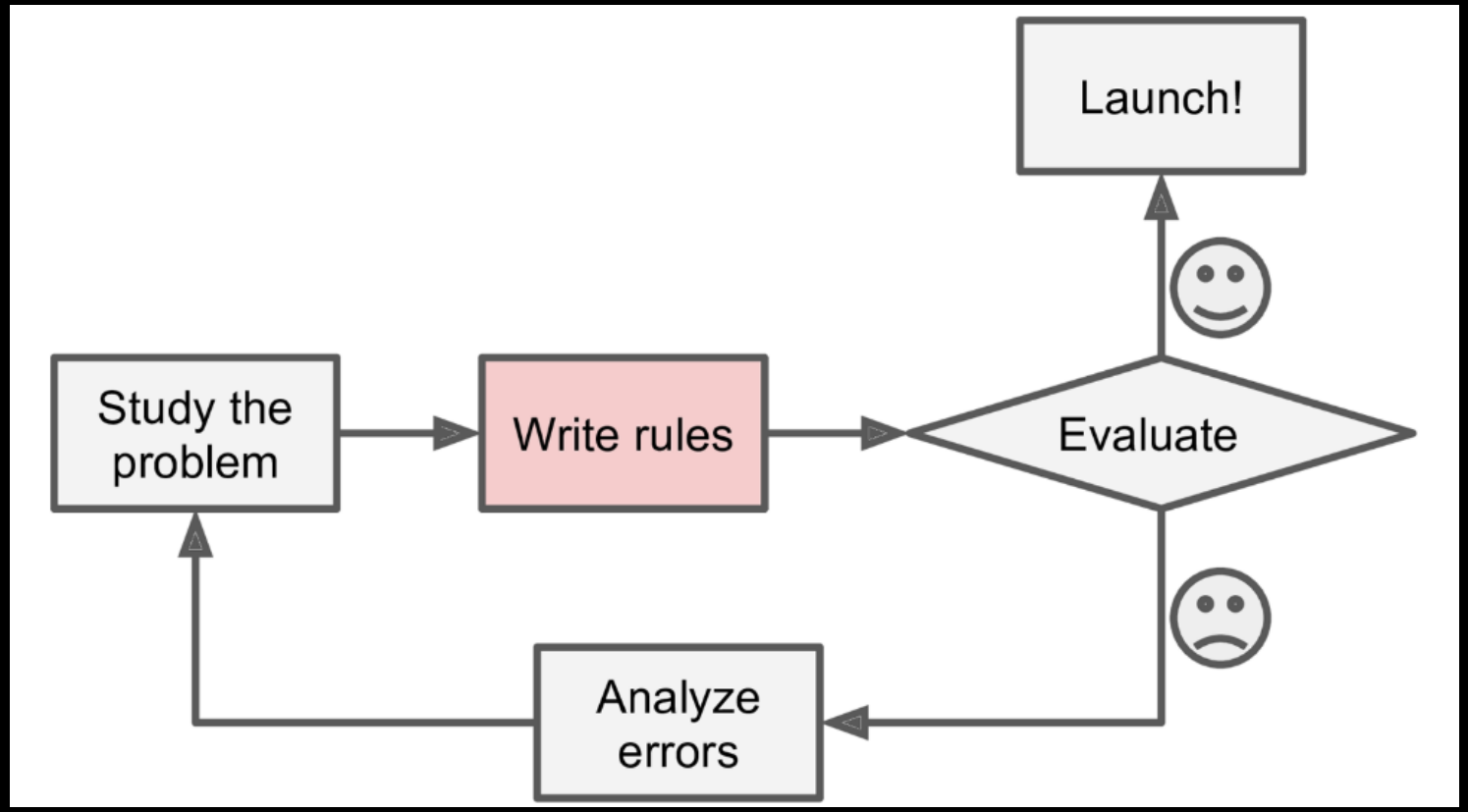
AI researcher Arthur Samuel coins the term "machine learning" as:

"Field of study that gives computers the ability to learn without being explicitly programmed."



Motivation for Machines that “Learn”

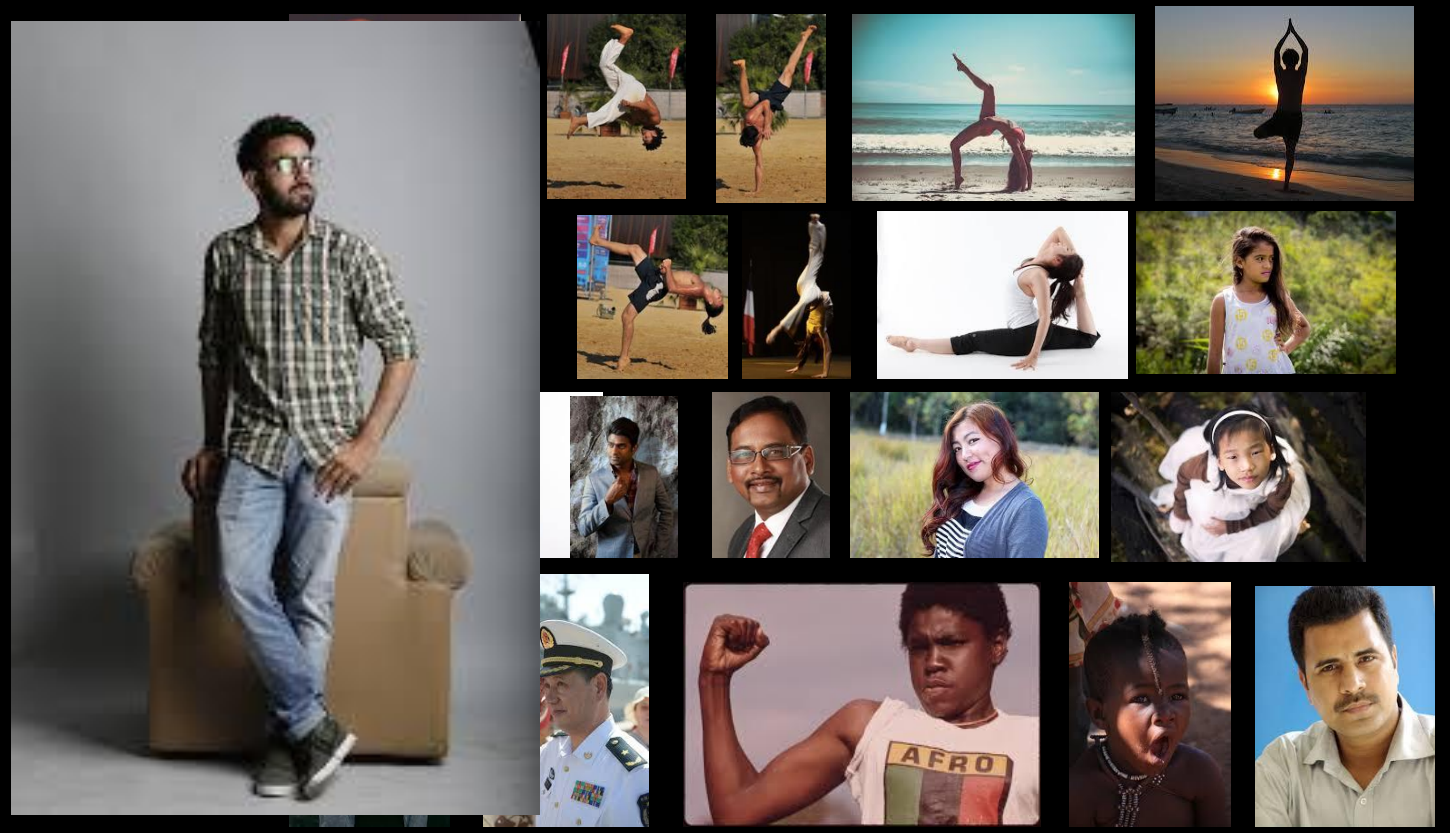
- Process for hand-crafted rules:



https://www.oreilly.com/library/view/hands-on-machine-learning/9781491962282/ch01.html

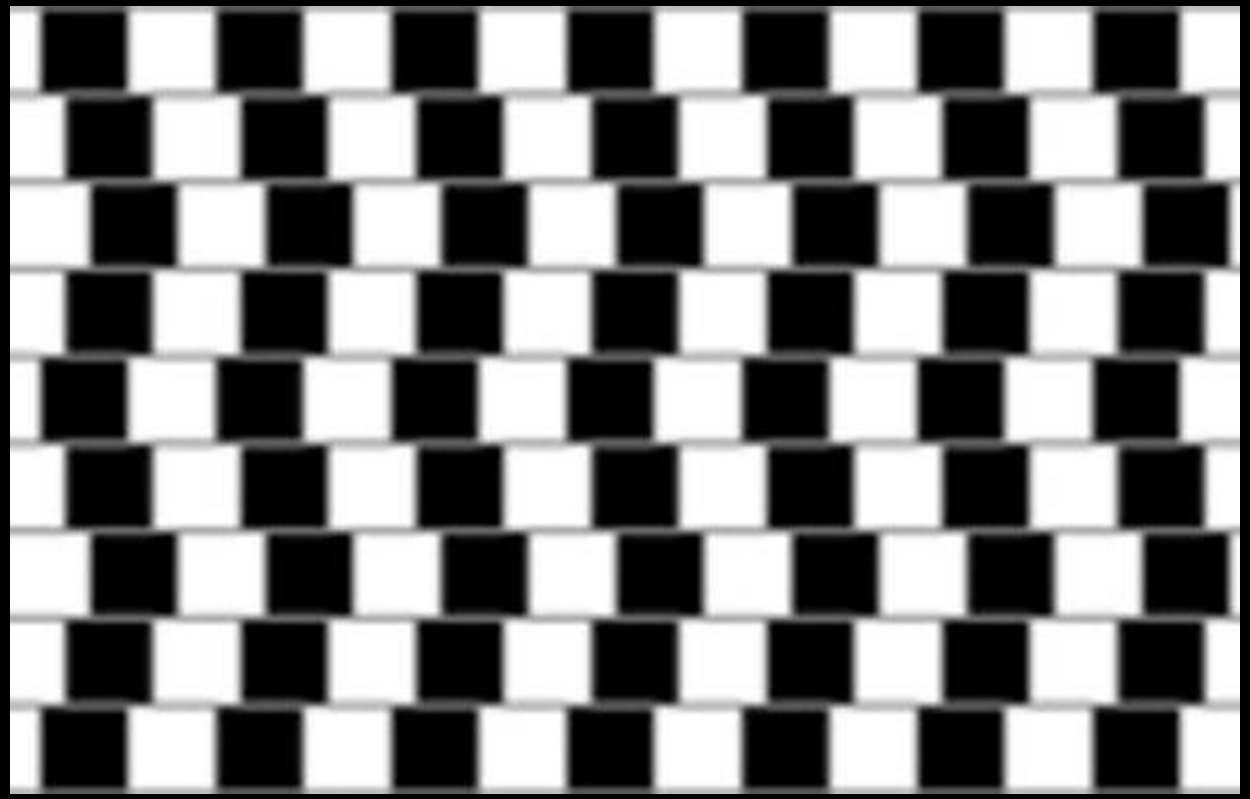
Motivation for Machines that “Learn”

e.g., What rules would you use to answer: “Is a person in the image?”



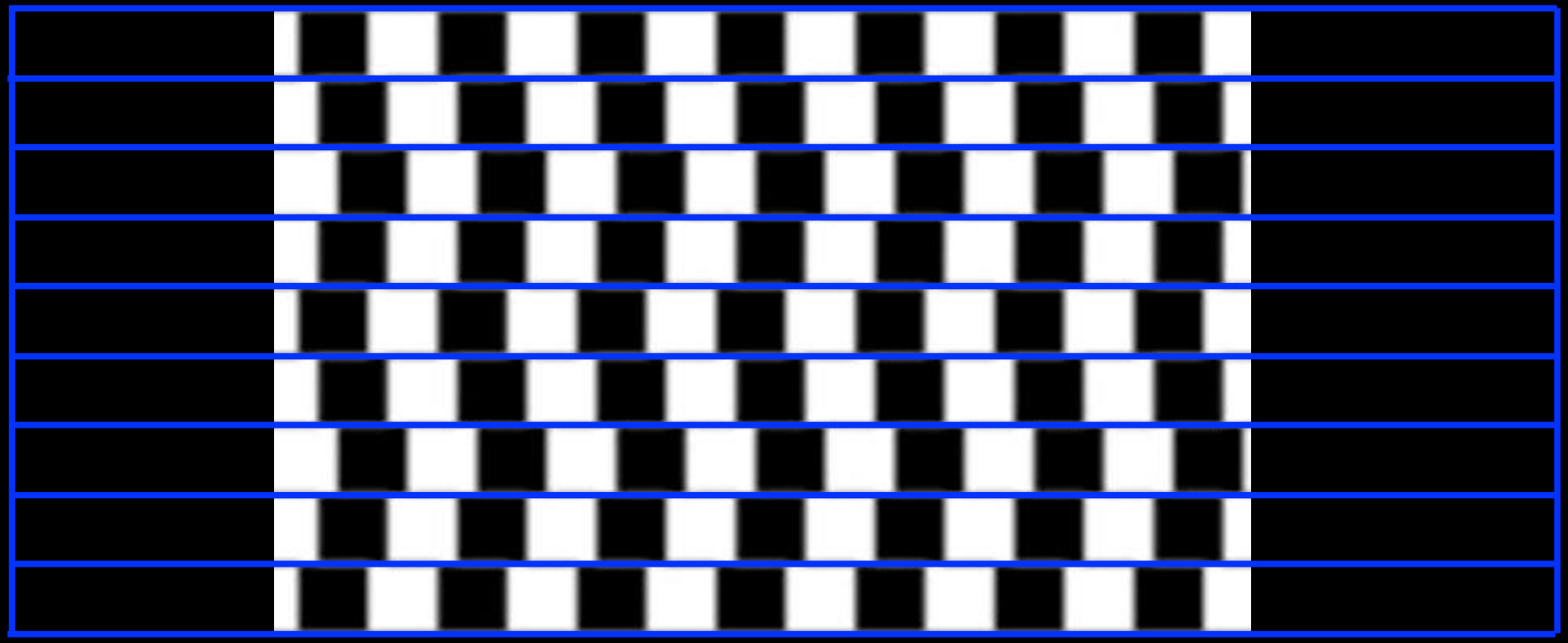
Motivation for Machines that “Learn”

e.g., are the horizontal lines parallel?



Motivation for Machines that “Learn”

e.g., are the horizontal lines parallel?



Motivation for Machines that “Learn”

1. It is hard to hand-craft a complete set of rules

2. We, as humans, may not devise the best rules for a machine since our brains (unconsciously) pre-process the data we sense

Learning Algorithm: ML Basics Recap

Developing machine learning models: choose data; e.g.,



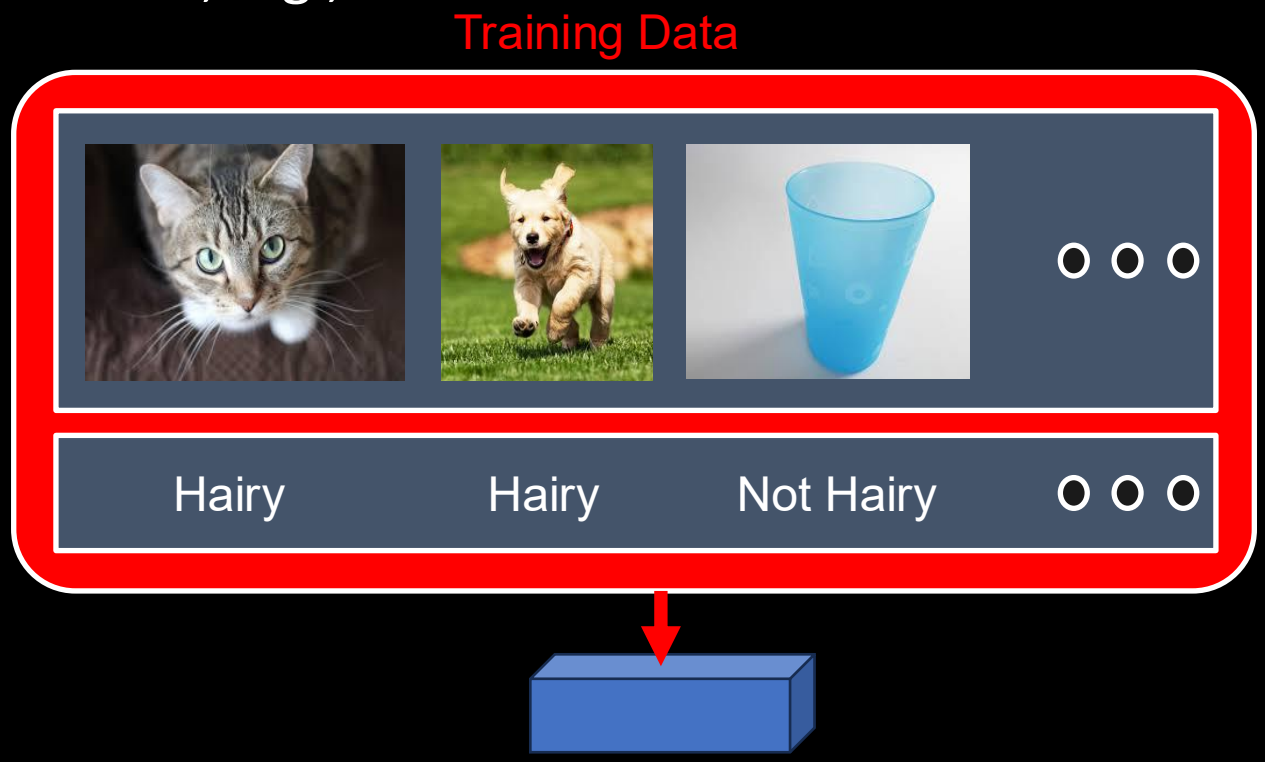
Learning Algorithm: ML Basics Recap

Developing machine learning models: choose data, split it; e.g.,



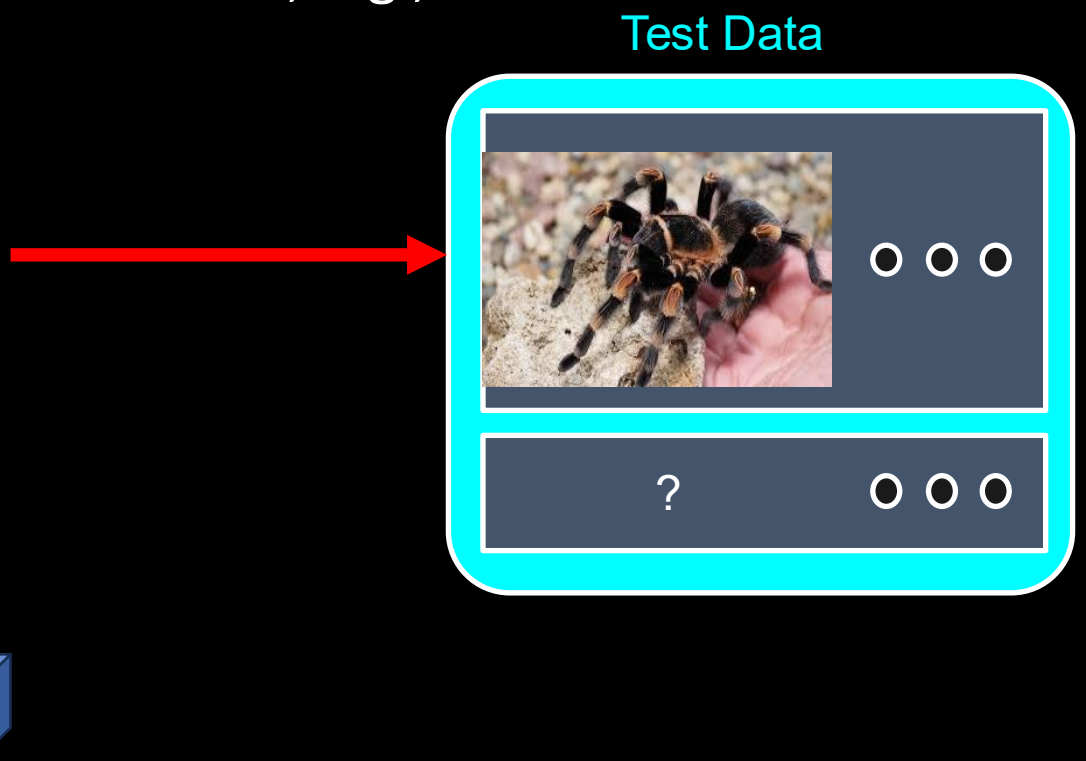
Learning Algorithm: ML Basics Recap

Developing machine learning models: choose data, split it, train model to minimize prediction error; e.g.,



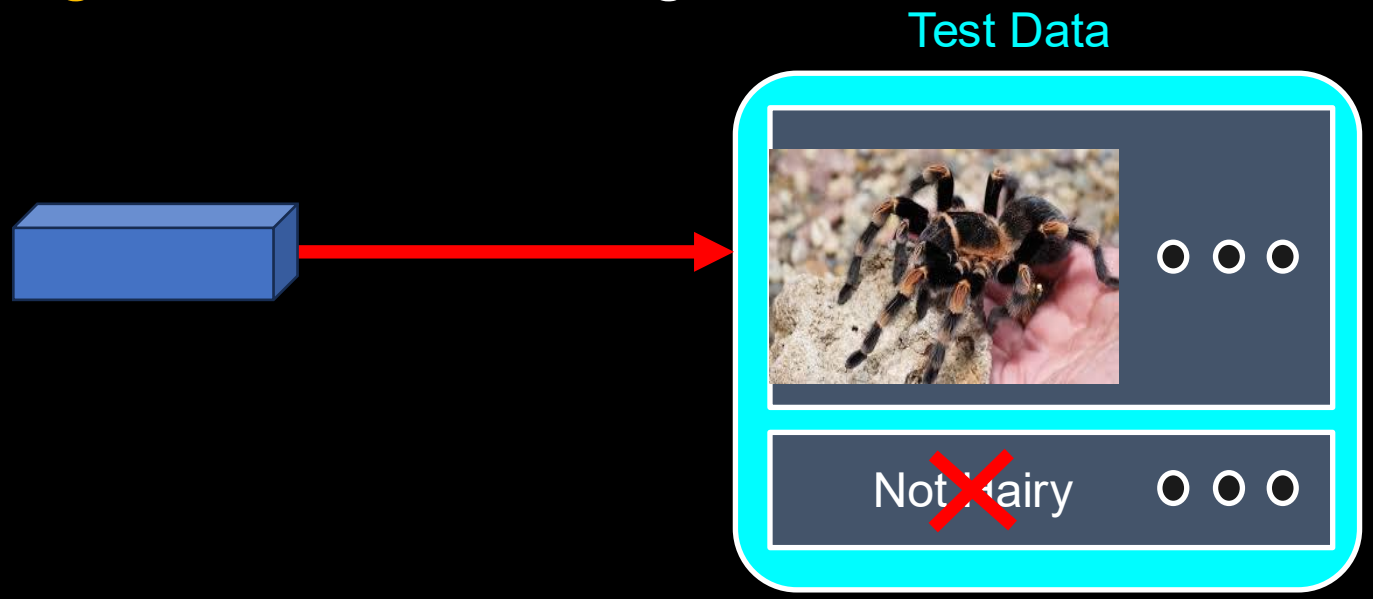
Learning Algorithm: ML Basics Recap

Developing machine learning models: choose data, split it, train model to minimize prediction error, and measure generalization error; e.g.,

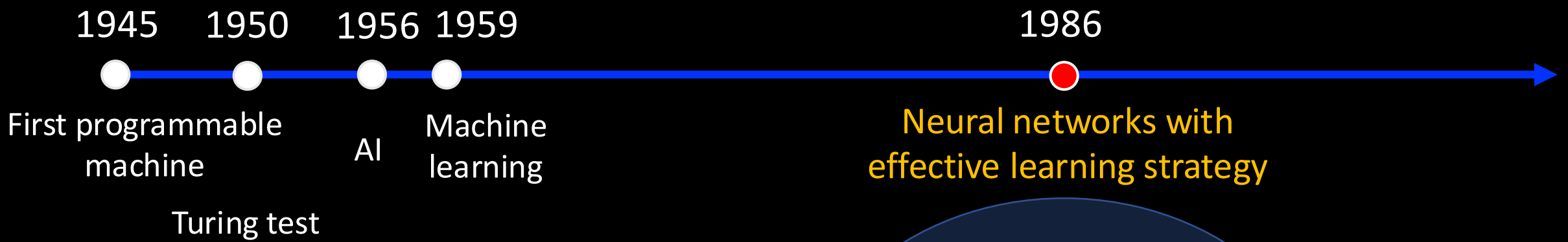


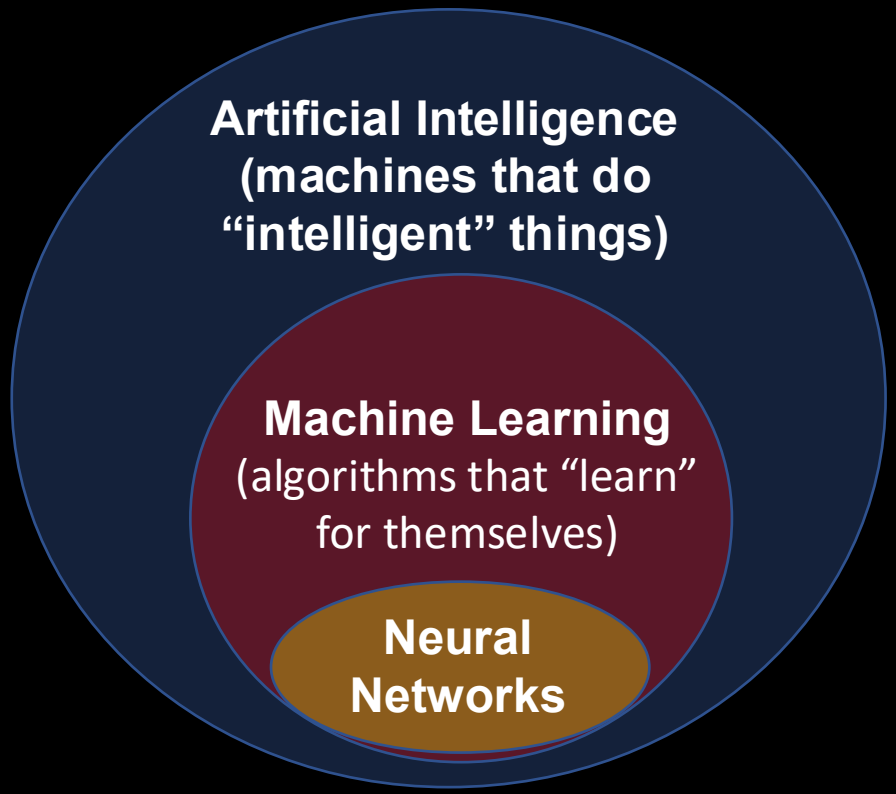
Learning Algorithm: ML Basics Recap

Developing machine learning models: choose data, split it, train model to minimize prediction error, and measure generalization error; e.g.,



Origins: Neural Networks with Deep Learning





Motivation for Neural Networks (NNs) Over Other Machine Learning (ML) Approaches

e.g., Is a person present?

INPUT



OTHER ML APPROACH: HAND-CRAFTED FEATURES

Feature Extraction



Prediction

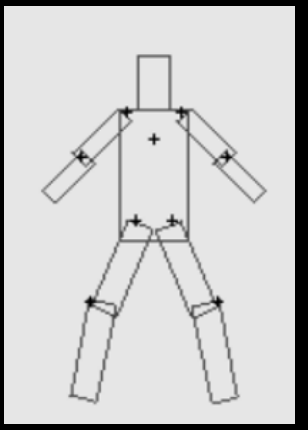


OUTPUT

Yes

Features that may help predict yes/no:

e.g., corners, lines, and model of expected body parts as connected shapes



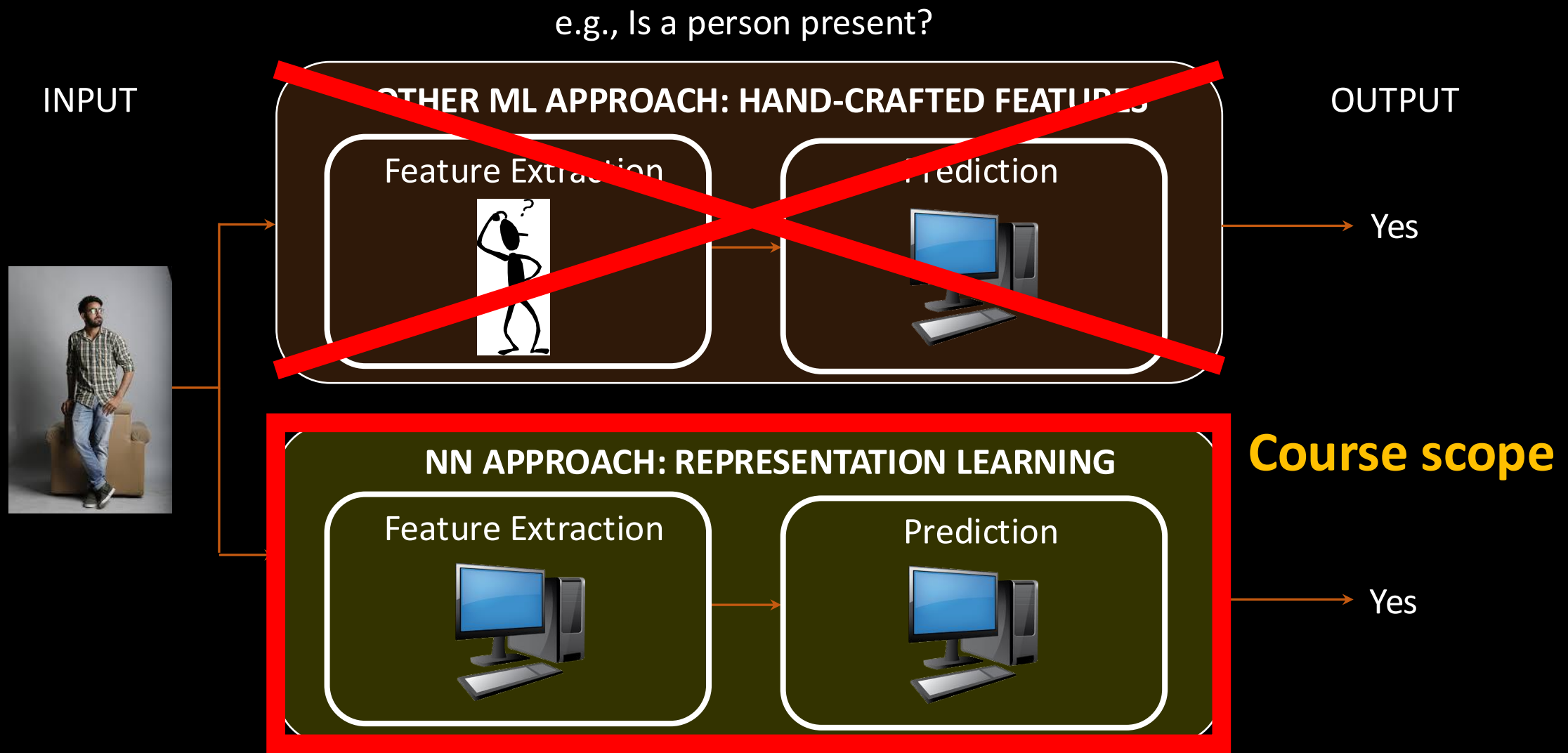
e.g., Pedro F Felzenszwalb and Daniel P Huttenlocher, IJCV 2004

Motivation for Neural Networks (NNs) Over Other Machine Learning (ML) Approaches



Challenging to hand-craft features!

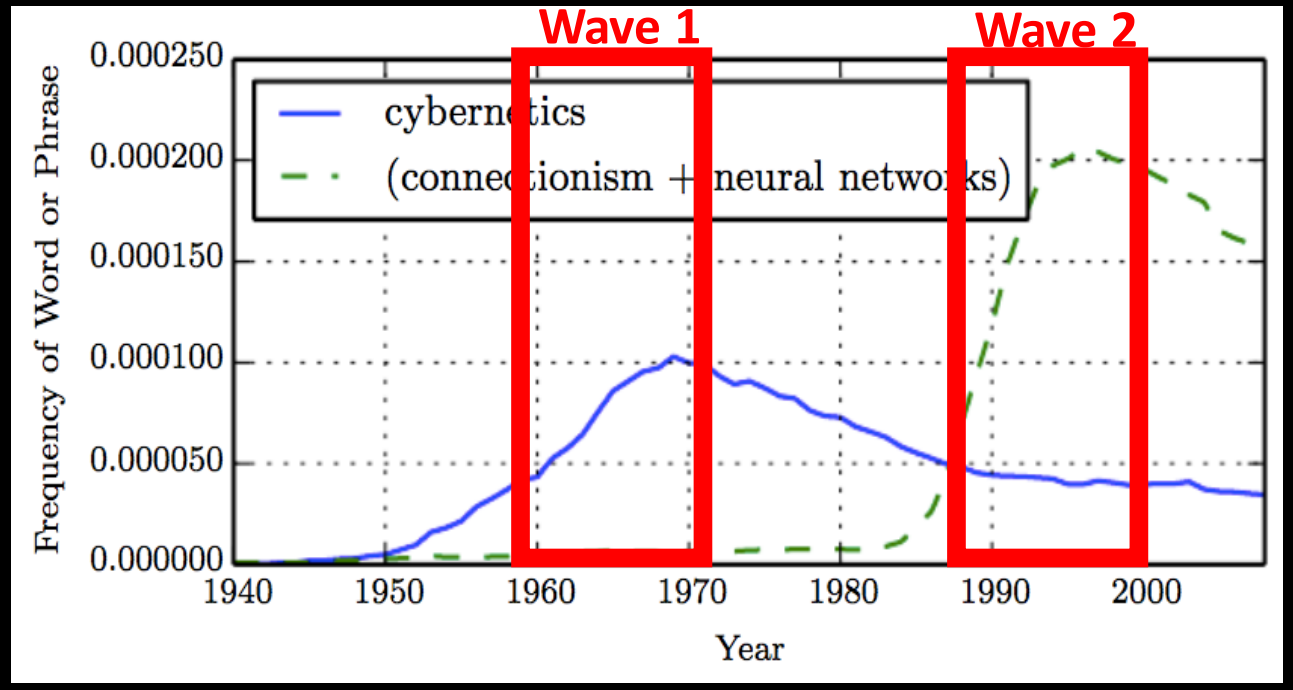
Motivation for Neural Networks (NNs) Over Other Machine Learning (ML) Approaches



Origins: Rises/Falls of Neural Network Popularity



Many names for the study of neural networks:



Ian Goodfellow, Yoshua Bengio, and Aaron Courville; Deep Learning, 2016

Why Are Neural Networks and Deep Learning So Popular Now? – Their Success in Practice!

Today's Topics

- Applications
- History of neural networks and deep learning
- How does a machine learn?
- Course logistics

Neural Networks: Key Ingredients for Success

An **architecture trained** on **data** with a **processor**
learns to extract useful patterns for prediction



Analogous to a Love Story of Partnering Up and Road Tripping Somewhere

Neural Networks: Key Ingredients for Success

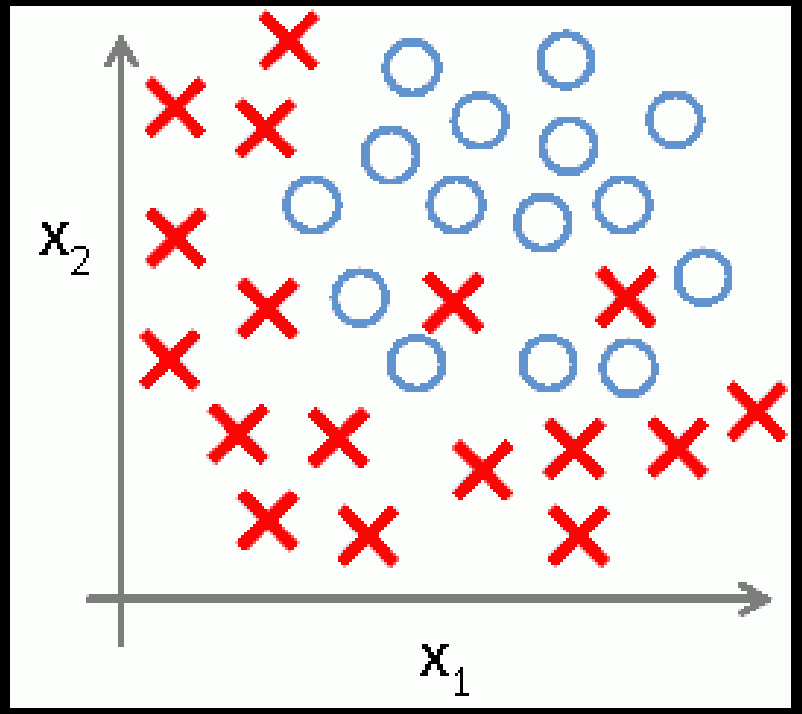
An **architecture trained** on data with a processor
learns to extract useful patterns for prediction



Analogous to a Love Story of Partnering Up and Road Tripping Somewhere

Architecture Design: Model-Based

- e.g., create model to separate x from o

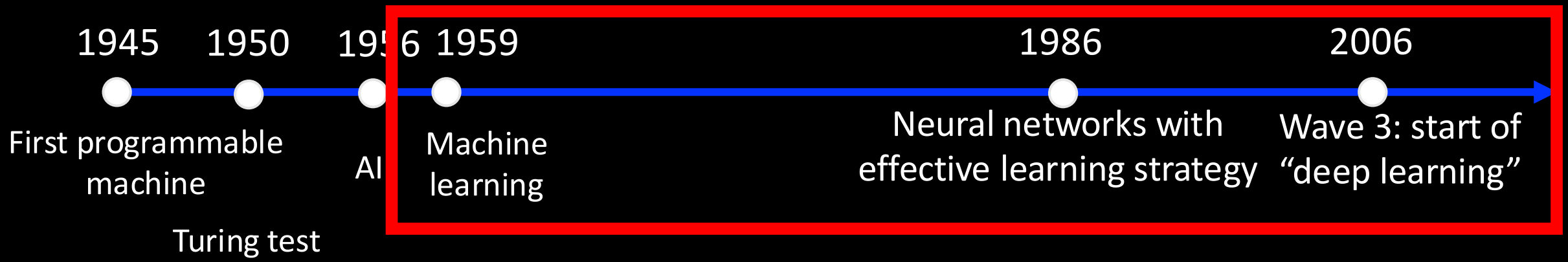


Class volunteer:
1) Draw a straight line (linear equation)
2) Draw a parabola (quadratic equation)
3) Draw any curve

Models with increasing representational capacity

https://medium.com/greyatom/what-is-underfitting-and-overfitting-in-machine-learning-and-how-to-deal-with-it-6803a989c76

Architecture Scope for Course: Last 70 Years



We will cover the history because progress has been cumulative, with each breakthrough extending the one before it

Neural Networks: Key Ingredients for Success

An architecture trained on **data** with a processor
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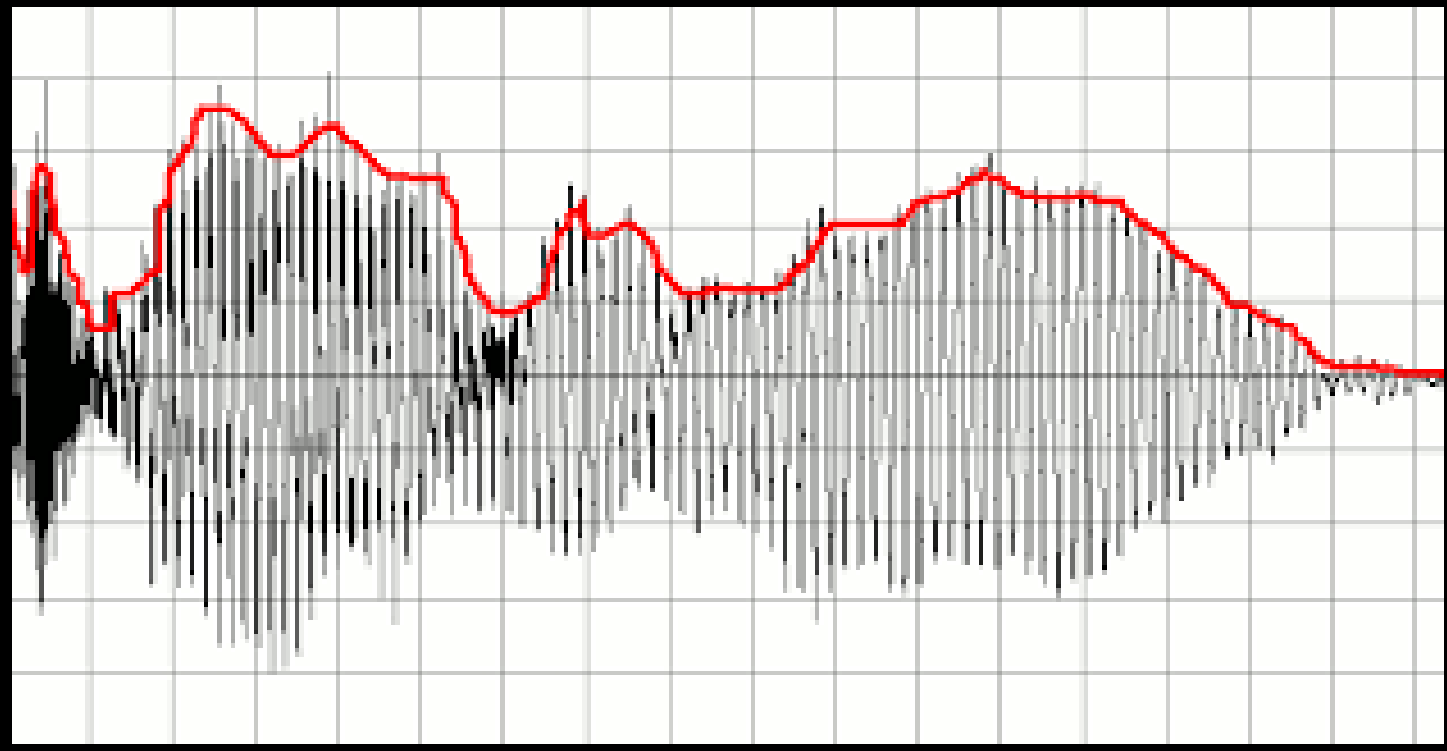
Data Types: What a Machine Learns From?



- Audio
  - Input?



e.g.,

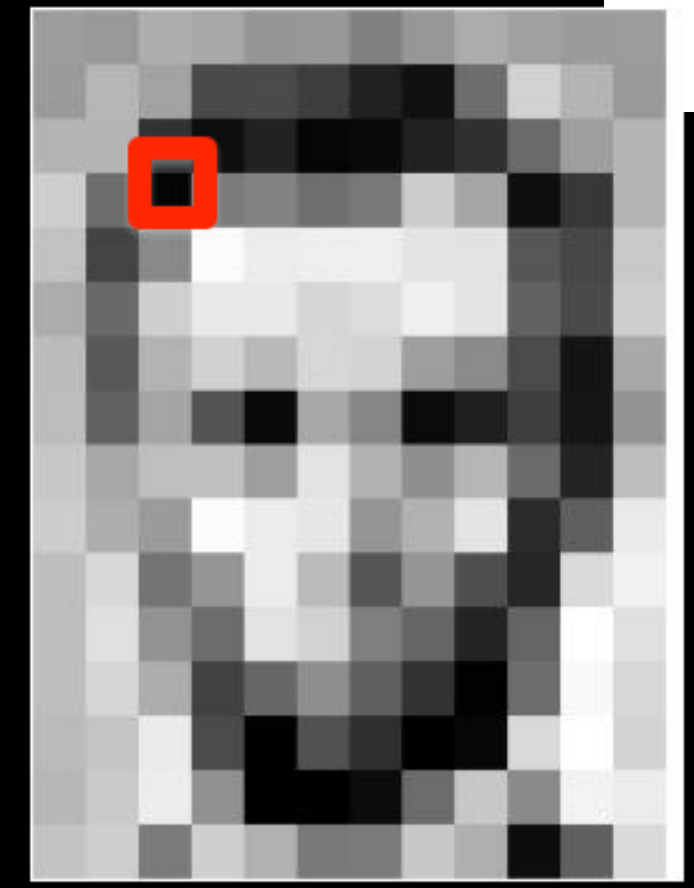


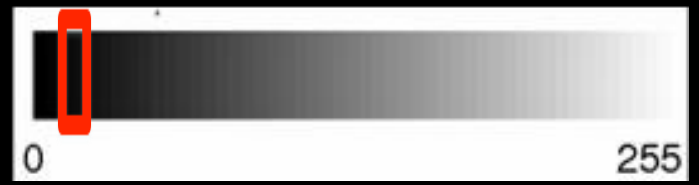
Data Types: What a Machine Learns From?



- Audio
  - Input?
- Images
  - Input?

157	153	174	168	150	152	129	151	172	161	155	156
155	182	163	74	75	62	33	17	110	210	180	154
180	180	50	14	34	6	10	33	48	106	159	181
206	109	5	24	131	111	120	204	166	15	56	180
194	68	137	251	237	239	239	228	227	87	71	201
172	105	207	233	233	214	220	239	228	98	74	206
188	88	179	209	185	215	211	158	139	75	20	169
189	97	165	84	10	168	134	11	31	62	22	148
199	168	191	193	158	227	178	143	182	106	36	190
205	174	155	252	236	231	149	178	228	43	95	234
190	216	116	149	236	187	86	150	79	38	218	241
190	224	147	108	227	210	127	102	36	101	255	224
190	214	173	66	103	143	96	50	2	109	249	215
187	196	235	75	1	81	47	0	6	217	255	211
183	202	237	145	0	0	12	108	200	138	243	236
195	206	123	207	177	121	123	200	175	13	96	218



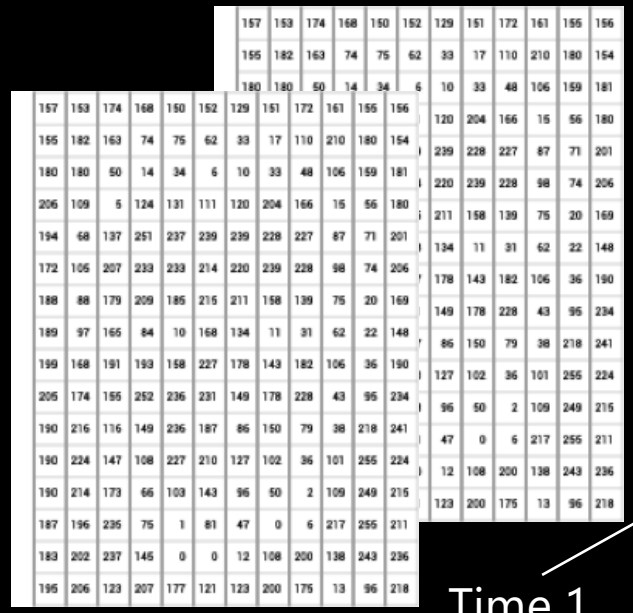


https://ai.stanford.edu/~syyeung/cvweb/tutorial1.html

Data Types: What a Machine Learns From?



- Audio
  - Input?
- Images
  - Input?
- Video
  - Input?

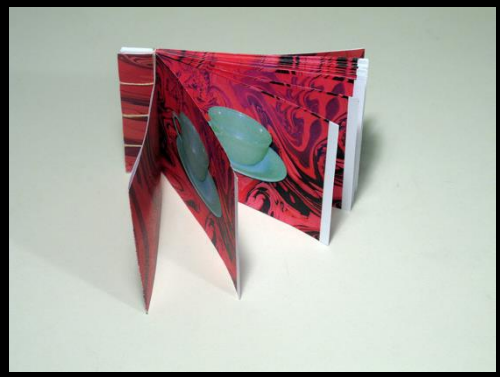


Time 1



1 hour

Analogous to:



Data Types: What a Machine Learns From?



- Audio
  - Input?
- Images
  - Input?
- Video
  - Input?
- Text
  - Input?

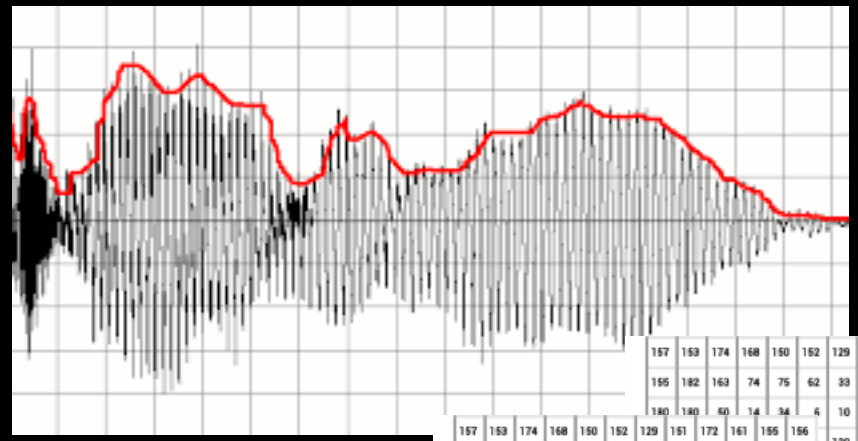
e.g.,

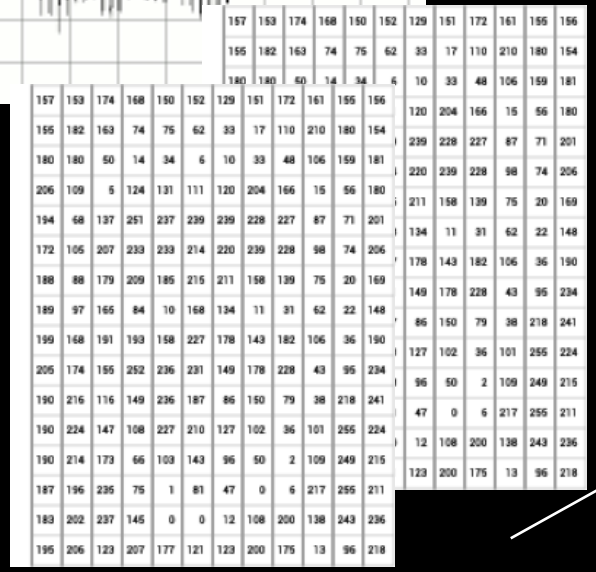


Data Types: What a Machine Learns From?

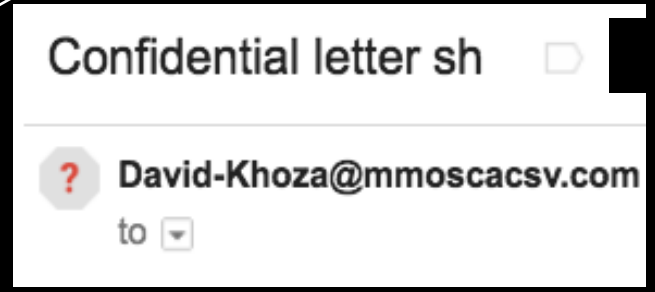


- Audio
  - Input?
- Images
  - Input?
- Video
  - Input?
- Text
  - Input?
- And more modalities
- Multi-modal: combines above









Data Types: Many Public Datasets Available

- Dataset creation is beyond the scope of this class
- We will benefit from other people's efforts:
  - Google Dataset Search
  - Kaggle datasets
  - Amazon's AWS datasets
  - Wikipedia's list
  - UC Irvine Machine Learning Repository
  - Quora.com
  - Reddit
  - Dataportals.org
  - Opendatamonitor.eu

Neural Networks: Key Ingredients for Success

An architecture trained on data with a **processor**
learns to extract useful patterns for prediction



Analogous to a Love Story of Partnering Up and Road Tripping Somewhere

Neural Networks: Key Ingredients for Success

e.g., train models using
GPUs (think Porsche) instead of CPUs (think Golf Cart)







Summary

An **architecture trained** on **data** with a **processor**
learns to extract useful patterns for prediction



Analogous to a Love Story of Partnering Up and Road Tripping Somewhere

Key Challenge: Where Will You Go?



https://www.theverge.com/2015/7/1/8880363/google-apologizes-photos-app-tags-two-black-people-gorillas





http://content.time.com/time/business/article/0,8599,1954643,00.html

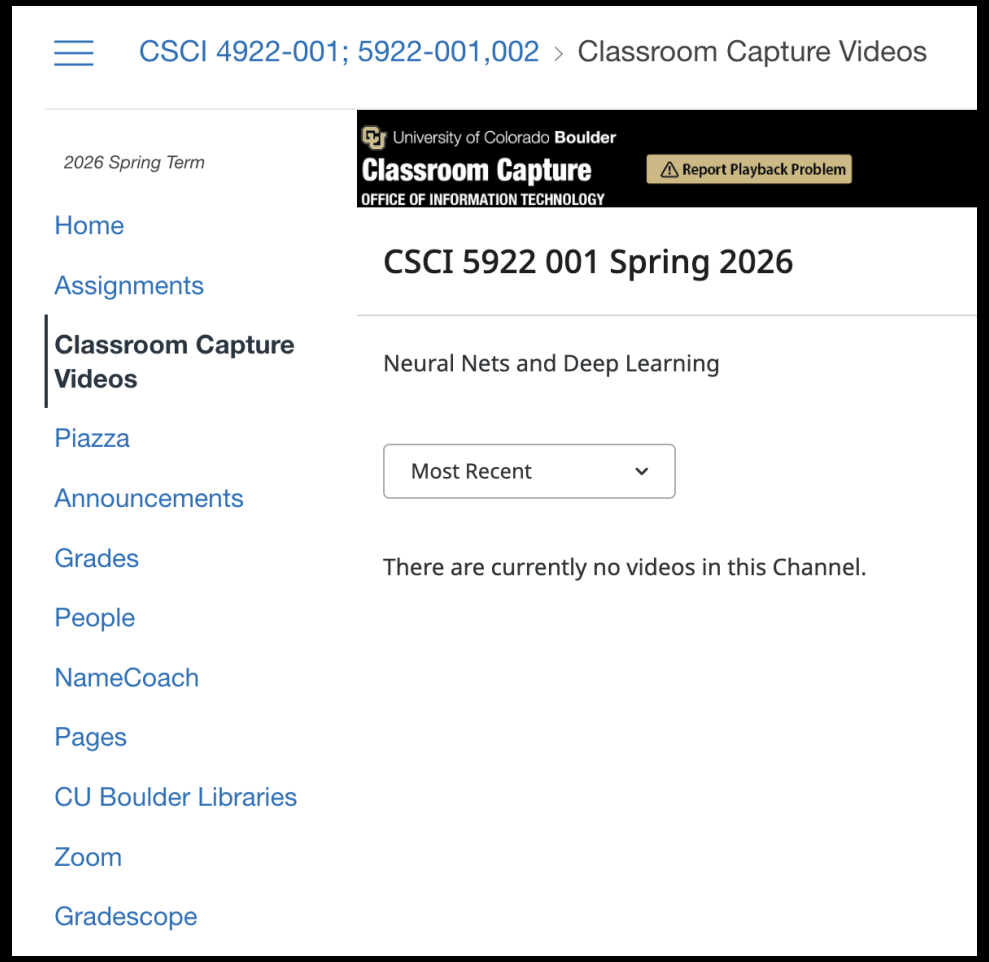
"Two kids bought their mom a camera for Mother's Day… when they took portrait pictures of each other, a message flashed across the screen asking, `Did someone blink?'"

Today's Topics

- Applications
- History of neural networks and deep learning
- How does a machine learn?
- Course logistics

Course Resources

- Website with Syllabus:
  - https://dannagurari.colorado.edu/course/neural-networks-and-deep-learning-spring-2026/
- Lecture Slides: hyperlinked from course website
- Lecture Recordings: available on Canvas immediately after each lecture



Course Objectives (1 of 5)

- Recognize and analyze common neural network architectures.
  Towards this aim, students will:
  - Identify popular architectures — including fully connected, recurrent, convolutional, and transformer-based networks — and implement their core mathematical operations by hand
  - Compare and contrast their design principles as well as their strengths and limitations for various tasks

Course Objectives (2 of 5)

- Recognize and analyze common techniques for training neural networks. Towards this aim, students will:
  - Identify key challenges with training neural networks of various depths and capacities
  - Compare and contrast techniques for overcoming learning challenges including training from scratch as well as performing transfer learning with frozen pretrained features, fine-tuning, and parameter-efficient methods (e.g., LoRA)

Course Objectives (3 of 5)

- Gain hands-on experience with neural networks and deep learning for various artificial intelligence tasks. Towards this aim, students will:
  - Implement basic neural networks and learning algorithms from scratch
  - Design, train, deploy, and evaluate neural networks models with multiple data types using the deep learning library PyTorch and modern hardware (GPUs)

Course Objectives (4 of 5)

- Explore responsible AI practices to both reduce environmental harm and promote ethical behavior in AI systems. Towards this aim, students will:
  - Experiment with sustainable techniques for model development including to speed up model inference and learning
  - Identify unethical behaviors of modern models, technical strategies to mitigate those behaviors, and societal frameworks to promote models' ethical use

Course Objectives (5 of 5)

- Apply what is learned to propose and critique novel projects involving neural networks and deep learning. Towards this aim, students will:
  - Propose a project idea involving neural networks and deep learning
  - Design and execute experiments or analyses supporting the proposed idea
  - Create a presentation about the project
  - Write a report about the project
  - Review fellow students' project presentations and provide constructive feedback

Course Schedule

Artificial Neurons (lecture 2)

Classical Neural Networks and Shallow Learning (lectures 3-6)

- Gradient Descent
- Fully Connected Feedforward, Recurrent, and Convolutional Neural Networks

Deep Learning and Transformers (lectures 7-19)

- Residual Learning, Batch Normalization, Hardware Acceleration, Fine Tuning, Big Data
- Applications with Visual, Textual, Audio, and Multimodal Data

Neural Networks in Society (lectures 20-26)

- Sustainable, Responsible, and Future Models
- Industry Perspectives

Three key breakthroughs:
"A spark can become a flame, a flame a fire."
- Kathryn Lasky







Q&A: “How does this course differ from other courses?”

- Other courses also cover neural networks and deep learning; e.g.,
  - Computer vision
  - Natural language processing
  - Reinforcement learning
- Unique benefit of this course:
  - Focus is on the fundamentals

Q&A: “Do I have the appropriate pre-requisites/background?”

- Programming competency
- Experience with probability/statistics and linear algebra
- Basic understanding of machine learning

Q&A: “What are required textbooks?”

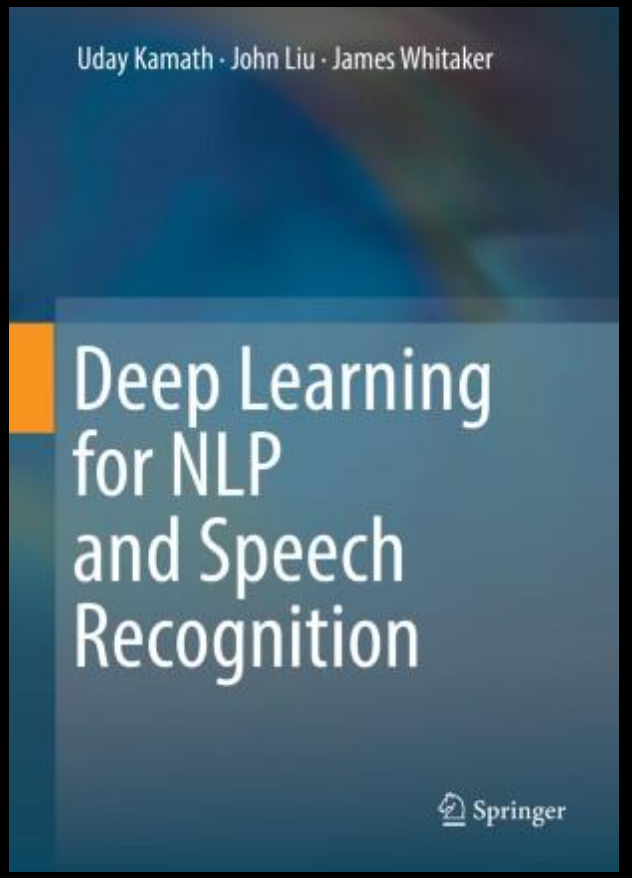
(available online for free)

Neural Networks and Deep Learning

Neural Networks and Deep Learning is a free online book. The book will teach you about:

- Neural networks, a beautiful biologically-inspired programming paradigm which enables a computer to learn from observational data
- Deep learning, a powerful set of techniques for learning in neural networks

(available online for free when connected to CU Boulder’s network or VPN)



Q&A: “What are the assignments?”

- 3 lab assignments (first assignment due in ~2 weeks)
- 3 problem sets
- Final project
  - Outline
  - Presentation
  - Peer evaluation
  - Final report
- Importantly: follow the assigned readings posted to the course website

Q&A: “How is my final grade determined?”

	% of Final Class Grade
Problem Sets	25%
Lab Assignments	45%
Final Project	30%

Q&A: “What can I expect for grading?”

- Timeline
  - Due to many students (~200), grades will be released up to 5 days after submission
- Policies:
  - Late policy: penalty of 1% of the grade per hour up to 12 hours, rounded up to the nearest hour; then 25% for up to 24 hours late; and then 50% for up to 48 hours late. After 48 hours, no credit will be given.
  - Regrade requests: must submit to TA within 2 weeks of receiving the grade

Introductions

Instructor: Danna Gurari; aka, Dr. G (she/her)

Teaching assistants (will hold office hours):



Everley Tseng (she/her)



Nick Cooper (he/him)



Jarek Reynolds (he/him)



Nolan Brady (he/him)

Q&A: “How Do I Get Answers to my Questions?”

- **Questions for Instructor**: I will stay after each class lecture to answer questions.
- **Piazza:** We encourage you to first post any questions to Piazza. This approach can benefit other students who have similar questions, as they can view the answers on Piazza. It also provides an opportunity for peers to answer your questions.
- **Office Hours**: The TAs will host office hours every business day of the week at times posted on the course website and Canvas. Instructions for how to attend are provided on the home page of Canvas.
- **Appointments**: If you have concerns that cannot be addressed during regular office hours, please email either Everley Tseng or Nick Cooper (course TAs) to request an appointment. They will check email Monday to Friday between 9am and 4pm. You can expect a response within one business day; if you do not receive a reply within that window, feel free to send a follow-up message. The TAs will involve the instructor for any items they are unable to address.

Q&A: “How Do I Get Involved with Research?”

- At the end of course, you will receive an announcement about how to apply to participate in research (with myself and my peers)

My Experience

2007-2010

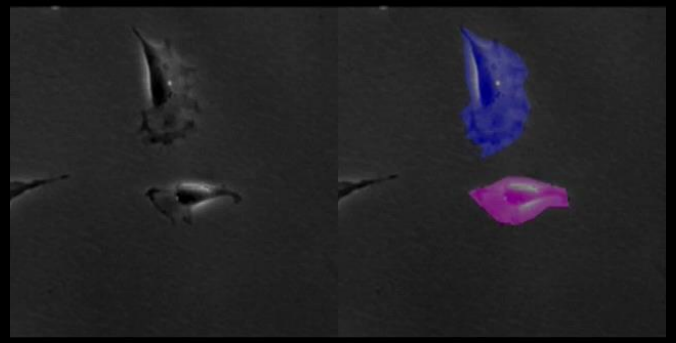
Software developer and project manager helping to record and analyze visible and infrared video



Source: Boulder Imaging

2010-2015

PhD student designing computer vision methods to segment and track cells in images and videos



2015-2017

Postdoctoral fellow conducting research projects related to vision and language problems



Is it edible or poisonous?

2017-Present

Assistant professor overseeing research projects related to many vision and language problems

e.g., image classification, object detection, semantic segmentation, object tracking, image captioning, visual question answering, style transfer, image inpainting, and image search

My Experience

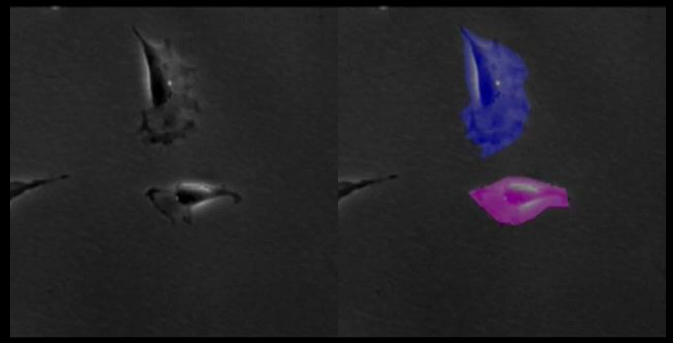
2007-2010	2010-2015	2015-2017	2017-Present

Software developer and project manager helping to record and analyze visible and infrared video



Source: Boulder Imaging

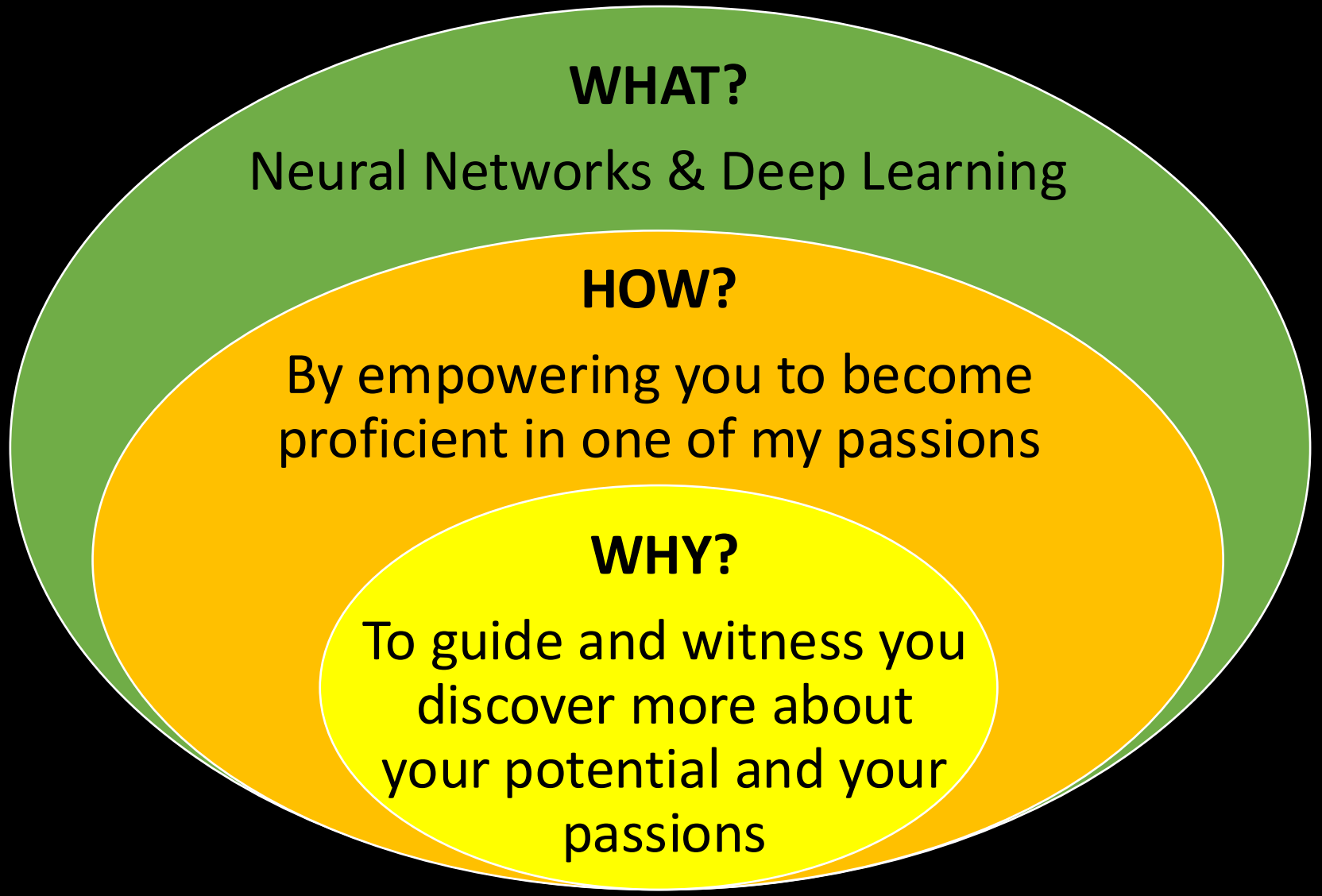
PhD student designing computer vision methods to segment and track cells in images and videos



41 publications involving deep learning

poisonous?

What is My “Why” for Teaching You…



Today's Topics

- Applications
- History of neural networks and deep learning
- How does a machine learn?
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The End